AS · Unit 1 · WCH11 Structure, Bonding and Introduction to Organic Chemistry

Formulae, equations and amounts of substance

Atoms are far too small to count one at a time, so chemists count them by weighing, by measuring gas volumes and by measuring solution volumes. This topic builds the idea of the mole, the three equations that link it to things you can measure, and the balanced equations that tell you how much of each substance reacts and forms. Almost every calculation in A level chemistry rests on what is here.

1. The mole and the Avogadro constant

A grain of salt holds about 10^{18} ions, so counting particles directly is hopeless. Instead chemists use a counting unit, the **mole** (symbol mol), in the same way that a baker counts eggs in dozens. One mole of anything contains the same number of particles: the **Avogadro constant**, $L = 6.02 \times 10^{23}\ \text{mol}^{-1}$.

The number was chosen so that the mass of one mole of a substance in grams is numerically the same as its relative mass. One mole of carbon-12 atoms has a mass of exactly 12 g, one mole of water molecules (relative mass 18.0) has a mass of 18.0 g, and so on. That is why weighing is a way of counting.

DEFINITION

Amount of substance is measured in moles. One mole is the amount of substance that contains as many particles as there are atoms in exactly 12 g of carbon-12, which is 6.02×10^{23} particles.

- **Relative atomic mass, A_r:** the weighted mean mass of an atom of an element, measured relative to one twelfth of the mass of an atom of carbon-12. It has no units.
- **Relative molecular mass, M_r:** the mean mass of a molecule relative to one twelfth of the mass of a carbon-12 atom. Add up the A_r values of every atom in the formula. For ionic compounds, which have no molecules, the same sum is called the **relative formula mass**.
- **Molar mass, M:** the mass of one mole of a substance, in g mol^{-1}. It has the same number as M_r: for $CaCO_3$, $M = 40.1 + 12.0 + 3 \times 16.0 = 100.1\ \text{g mol}^{-1}$.
- **Number of particles:** $N = n \times L$, where n is the amount in moles.

Always say which particle you are counting. One mole of water molecules contains two moles of hydrogen atoms, one mole of oxygen atoms and three moles of atoms altogether. One mole of sodium chloride contains one mole of Na^+ ions and one mole of Cl^- ions, so two moles of ions in total.

Counting atoms from a mass

1. How many atoms are there in 5.00 g of aluminium? (A_r: Al = 27.0)
2. Find the amount: $n = \dfrac{m}{M} = \dfrac{5.00}{27.0} = 0.1852\ \text{mol}$

   Keep the unrounded value in your calculator for the next step.
3. Multiply by the Avogadro constant: $N = 0.1852 \times 6.02 \times 10^{23} = 1.11 \times 10^{23}$ atoms

Atoms inside molecules

1. How many atoms are there in 0.500 mol of ammonia, NH_3?
2. Number of molecules: $0.500 \times 6.02 \times 10^{23} = 3.01 \times 10^{23}$
3. Each molecule contains 4 atoms (one N and three H), so the number of atoms is $4 \times 3.01 \times 10^{23} = 1.20 \times 10^{24}$.

The same idea gives the mass of a single atom: one carbon-12 atom has a mass of $12 \div (6.02 \times 10^{23}) = 1.99 \times 10^{-23}$ g.

2. Amount from mass

For any pure solid, liquid or gas, the amount in moles comes from its mass and its molar mass:

$$n = \frac{m}{M} \qquad \text{amount (mol)} = \frac{\text{mass (g)}}{\text{molar mass (g mol}^{-1}\text{)}}$$

The mass must be in grams. Convert first: 1 g = 1000 mg, 1 kg = 1000 g and 1 tonne = 10^6 g. Rearranging gives $m = n \times M$ and $M = m/n$, the last of which is how you identify an unknown substance from a measured mass and amount.

This page uses these relative atomic masses from the periodic table: H 1.0, C 12.0, N 14.0, O 16.0, Na 23.0, Mg 24.3, Al 27.0, S 32.1, Cl 35.5, K 39.1, Ca 40.1, Fe 55.8, Cu 63.5.

EXAM TIP

Give your final answer to the same number of significant figures as the least precise piece of data in the question, usually 3. Never round in the middle of a calculation: carry the full calculator value through and round only at the end, otherwise rounding errors pile up.

Hydrated salts and water of crystallisation

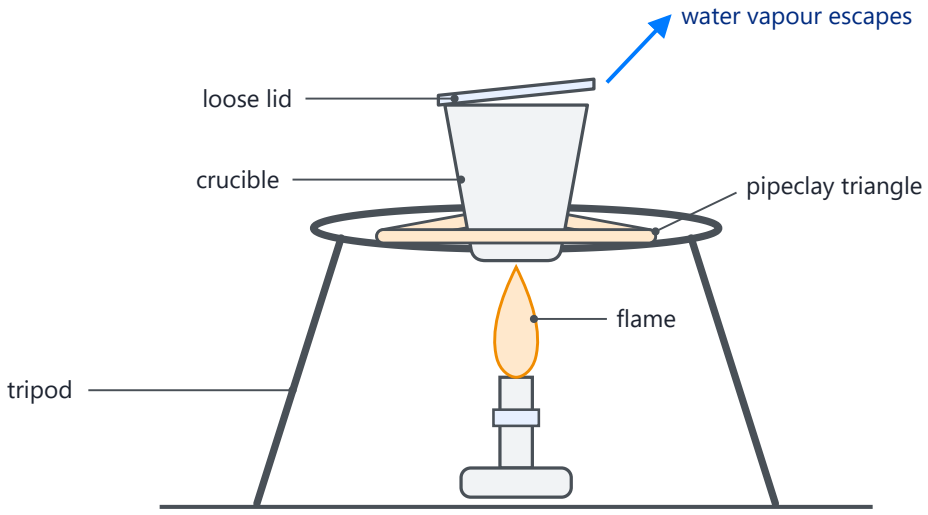
Many salts crystallise from water with water molecules built into their crystal lattice. This is **water of crystallisation**, and the formula shows it after a dot. Blue hydrated copper(II) sulfate is $CuSO_4 \cdot 5H_2O$. Heating drives the water off and leaves white anhydrous $CuSO_4$. The water is part of the compound, so it counts in the molar mass.

Amount of a hydrated salt

1. What amount of $CuSO_4 \cdot 5H_2O$ is in 2.50 g of the blue crystals?
2. $M = 63.5 + 32.1 + (4 \times 16.0) + (5 \times 18.0) = 249.6\ \text{g mol}^{-1}$

   Each H_2O has $M = 18.0$. Forgetting the water, and using 159.6, is the usual error.
3. $n = \dfrac{2.50}{249.6} = 0.0100\ \text{mol}$

To find the number of water molecules, x, in a hydrated salt, heat a weighed sample in a crucible to drive off the water, then weigh the anhydrous solid that is left. The same method measures the mass lost in other thermal decompositions, or the mass gained when a metal such as magnesium is burned to its oxide.



A crucible on a pipeclay triangle. The lid stops solid spitting out, but it sits loosely so that water vapour can escape.

- **Method:** weigh the clean, dry crucible and lid; add about 2 g of the hydrated salt and weigh again; heat strongly for a few minutes; let it cool and reweigh; then reheat, cool and reweigh until two readings agree. This is **heating to constant mass**, and it shows that all the water has gone.
- **Sample size:** about 2 g is right. A much larger sample may not lose all its water. A tiny sample gives a mass change so small that the balance's uncertainty becomes a large percentage of it.
- **Dry crucible:** any water already on the crucible is lost on heating and counted as water of crystallisation, so x comes out too big.
- **Cooling:** cool the anhydrous solid in a desiccator, because many anhydrous salts reabsorb water from the air.

Finding x in $MgSO_4 \cdot xH_2O$

1. 4.92 g of hydrated magnesium sulfate, $MgSO_4 \cdot xH_2O$, is heated to constant mass. 2.40 g of anhydrous $MgSO_4$ is left. Find x.
2. Mass of water lost $= 4.92 - 2.40 = 2.52$ g
3. $n(MgSO_4) = \dfrac{2.40}{120.4} = 0.01993$ mol and $n(H_2O) = \dfrac{2.52}{18.0} = 0.140$ mol

   $M(MgSO_4) = 24.3 + 32.1 + 4 \times 16.0 = 120.4\ \text{g mol}^{-1}$.
4. Divide by the smaller amount: $\dfrac{0.140}{0.01993} = 7.02$, so $x = 7$ and the formula is $MgSO_4 \cdot 7H_2O$.

COMMON MISTAKE

Think about which way an error pushes the answer. If the salt is not heated to constant mass, some water stays behind, the mass lost is too small and x comes out **too low**. If solid spits out of the crucible, the mass lost is too big and x comes out **too high**.

3. Empirical and molecular formulae

- **Empirical formula:** the simplest whole-number ratio of the atoms of each element in a compound.
- **Molecular formula:** the actual number of atoms of each element in one molecule. It is always a whole-number multiple of the empirical formula.

Ethene is C_2H_4 and cyclohexane is C_6H_{12}, but both have the empirical formula CH_2. Analysis of a compound tells you the mass (or percentage by mass) of each element, which only gives the ratio of atoms. You need the relative molecular mass, for example from a mass spectrum, to get the molecular formula.

1. Divide the mass (or percentage) of each element by its A_r. This gives the amount of each element in moles.
2. Divide every answer by the smallest one.
3. If any result is close to x.5, x.33 or x.25, multiply all of them by 2, 3 or 4 to reach whole numbers.
4. For the molecular formula, divide M_r by the empirical formula mass and multiply the empirical formula by the result.

Empirical and molecular formula from percentages

1. A hydrocarbon contains 85.7% carbon and 14.3% hydrogen by mass. Its M_r is 56.0. Find its molecular formula.
2. Take 100 g, so there are 85.7 g of C and 14.3 g of H. Divide by A_r: C: $85.7 \div 12.0 = 7.14$ mol; H: $14.3 \div 1.0 = 14.3$ mol
3. Divide by the smaller value: C: $7.14 \div 7.14 = 1$; H: $14.3 \div 7.14 = 2.00$. The empirical formula is CH_2.
4. Empirical formula mass $= 12.0 + 2.0 = 14.0$, and $56.0 \div 14.0 = 4$. The molecular formula is C_4H_8.

M_r does not need to be exact. The multiple must be a whole number, so a measured value of 55 or 57 still points to 4.

When the ratio is not a whole number

1. An oxide of iron contains 69.9% iron by mass. Find its empirical formula.
2. Oxygen makes up the rest: $100 - 69.9 = 30.1\%$. Fe: $69.9 \div 55.8 = 1.253$; O: $30.1 \div 16.0 = 1.881$
3. Divide by the smaller: Fe = 1, O = $1.881 \div 1.253 = 1.50$
4. 1.50 is not a whole number, so double both: Fe = 2, O = 3. The empirical formula is Fe_2O_3.

Rounding 1.50 to 2 would give FeO_2, which is wrong. Only round values that are within about 0.1 of a whole number.

4. Balanced equations and ionic equations

Atoms are not created or destroyed in a chemical reaction, so a balanced equation has the same number of each kind of atom on both sides. You balance it by changing the numbers in front of formulas, never the formulas themselves. The balancing numbers give the **mole ratio** in which substances react: in $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, 1 mol of nitrogen reacts with 3 mol of hydrogen to make 2 mol of ammonia. State symbols show the physical state: (s) solid, (l) liquid, (g) gas and (aq) dissolved in water.

Ionic equations

When ionic compounds dissolve, the ions separate and move independently. Many of them take no part in the reaction and are present unchanged at the end. These are **spectator ions**. An ionic equation leaves them out and shows only the particles that change, which reveals what is really happening. It must balance for charge as well as for atoms.

- **Split into ions:** soluble ionic compounds in solution (aq), including strong acids such as HCl(aq), H_2SO_4(aq) and HNO_3(aq), and soluble hydroxides such as NaOH(aq).

- **Do not split:** solids (s), including precipitates, insoluble oxides and carbonates; liquids such as water; gases; and covalent molecules.

Writing an ionic equation

1. Start with the balanced full equation, including state symbols:
   $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

2. Split every aqueous ionic compound into its ions:
   $Ag^+(aq) + NO_3^-(aq) + Na^+(aq) + Cl^-(aq) \rightarrow AgCl(s) + Na^+(aq) + NO_3^-(aq)$

   AgCl is a solid precipitate, so it stays together.

3. Cross out ions that appear unchanged on both sides (Na^+ and NO_3^-):
   $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$

   Check the charges: 1+ and 1− on the left give zero overall, and so does the neutral solid on the right. Any soluble silver salt mixed with any soluble chloride gives this same equation, which is why ionic equations are so useful.

Displacement reactions

A more reactive metal displaces a less reactive metal from a solution of its compound, because the more reactive metal gives up electrons more readily. Zinc dropped into blue copper(II) sulfate solution becomes coated in red-brown copper, the blue colour fades and the mixture warms up:

$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$ ionic: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

Halogens behave the same way: a more reactive halogen, which is a stronger oxidising agent, displaces a less reactive one from a solution of its halide ions. The colour change shows the new halogen forming.

Mix	Ionic equation	What you see
chlorine water + potassium bromide	$Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$	colourless solution turns yellow-orange
chlorine water + potassium iodide	$Cl_2(aq) + 2I^-(aq) \rightarrow 2Cl^-(aq) + I_2(aq)$	colourless solution turns brown
bromine water + potassium iodide	$Br_2(aq) + 2I^-(aq) \rightarrow 2Br^-(aq) + I_2(aq)$	orange solution turns brown
bromine water + potassium chloride	no reaction	stays orange: bromine is less reactive than chlorine

You will meet the reasons for these trends in the topic on redox and the halogens.

Reactions of acids

An **acid** releases hydrogen ions, $H^+(aq)$, in water. The common strong acids are hydrochloric acid HCl, sulfuric acid H_2SO_4 and nitric acid HNO_3. A **base** neutralises an acid: metal oxides, metal hydroxides, carbonates and ammonia are all bases. An **alkali** is a base that dissolves in water to release hydroxide ions, $OH^-(aq)$, for example NaOH, KOH and aqueous ammonia. A **salt** forms when the hydrogen of an acid is replaced by a metal ion or an ammonium ion.

Reaction	Full equation	Ionic equation	What you see
acid + metal → salt + hydrogen	$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$	$Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$	fizzing; the metal dissolves; the mixture warms
acid + alkali → salt + water	$NaOH(aq) + HNO_3(aq) \rightarrow NaNO_3(aq) + H_2O(l)$	$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$	no visible change; the temperature rises
acid + insoluble base → salt + water	$CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$	$CuO(s) + 2H^+(aq) \rightarrow Cu^{2+}(aq) + H_2O(l)$	black solid dissolves to give a blue solution
acid + solid carbonate → salt + water + carbon dioxide	$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$	$CaCO_3(s) + 2H^+(aq) \rightarrow Ca^{2+}(aq) + H_2O(l) + CO_2(g)$	effervescence; the solid dissolves; the gas turns limewater milky
acid + carbonate solution	$Na_2CO_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$	$CO_3^{2-}(aq) + 2H^+(aq) \rightarrow H_2O(l) + CO_2(g)$	effervescence
acid + ammonia → ammonium salt	$NH_3(aq) + HCl(aq) \rightarrow NH_4Cl(aq)$	$NH_3(aq) + H^+(aq) \rightarrow NH_4^+(aq)$	no visible change in solution

EXAM TIP

Every neutralisation of a strong acid by an alkali has the same ionic equation, $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$. In the copper(II) oxide reaction only the sulfate ion is a spectator: the oxide is a solid and the copper ends up in solution, so both change state and must appear.

Precipitation reactions

Mixing two solutions can bring together a cation and an anion that form an insoluble solid. The solid appears as a cloudy **precipitate**, and its colour often identifies the ions.

Ions mixed	Ionic equation	Precipitate
silver and chloride	$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$	white
silver and bromide	$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$	cream
silver and iodide	$Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$	pale yellow
barium and sulfate	$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$	white
copper(II) and hydroxide	$Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$	pale blue
iron(III) and hydroxide	$Fe^{3+}(aq) + 3OH^-(aq) \rightarrow Fe(OH)_3(s)$	orange-brown

Making salts

- **Soluble salt from an insoluble base, metal or carbonate:** add the solid to warm acid until no more dissolves, so that it is in excess and all the acid is used up. Filter off the excess solid. Evaporate the filtrate to about half its volume, then leave it to cool and crystallise slowly. Filter off the crystals and dry them between filter papers.

- **Soluble salt from an alkali:** both reactants are solutions, so excess cannot be filtered off. Do a titration with an indicator to find the exact volumes that react, then repeat with those volumes and no indicator, and crystallise as above.
- **Insoluble salt:** mix solutions containing its two ions, filter off the precipitate, wash it with distilled water to remove soluble impurities, and dry it.

The yield of crystals is always below 100%, because some salt stays dissolved in the solution left behind and some is lost on the filter paper and in transfers.

5. Concentration of solutions

A solution is a solute dissolved in a solvent, usually water. Its concentration tells you how much solute is in each cubic decimetre of solution:

$$c = \frac{n}{V} \qquad \text{concentration (mol dm}^{-3}) = \frac{\text{amount (mol)}}{\text{volume of solution (dm}^3)}$$

1 dm^3 is a cube 10 cm on each side, so it equals 1000 cm^3 (one litre), and 1 m^3 = 1000 dm^3. Volumes are usually measured in cm^3, so divide by 1000 before using the equation. Concentration can also be written in g dm^{-3}, the mass of solute in each dm^3: multiply the concentration in mol dm^{-3} by the molar mass.

Concentration of a standard solution

1. 3.18 g of anhydrous sodium carbonate, Na_2CO_3, is dissolved in water and made up to 250.0 cm^3. Find its concentration in mol dm^{-3} and in g dm^{-3}.
2. $M = (2 \times 23.0) + 12.0 + (3 \times 16.0) = 106.0\ \text{g mol}^{-1}$, so $n = \dfrac{3.18}{106.0} = 0.0300\ \text{mol}$
3. Volume in dm^3: $250.0 \div 1000 = 0.2500\ \text{dm}^3$
4. $c = \dfrac{0.0300}{0.2500} = 0.120\ \text{mol dm}^{-3}$
5. In g dm^{-3}: $0.120 \times 106.0 = 12.7\ \text{g dm}^{-3}$

   Check: 3.18 g in a quarter of a cubic decimetre is the same as 12.7 g in a whole one.

Concentrations of ions

A dissolved ionic compound separates into its ions, so the concentration of an ion can be larger than the concentration of the compound. In 0.200 mol dm^{-3} sodium sulfate, $Na_2SO_4(aq) \rightarrow 2Na^+(aq) + SO_4^{2-}(aq)$, so $[Na^+]$ = 0.400 mol dm^{-3} and $[SO_4^{2-}]$ = 0.200 mol dm^{-3}.

Counting ions in a solution

1. How many chloride ions are in 25.0 cm^3 of 0.150 mol dm^{-3} calcium chloride solution?
2. $n(CaCl_2) = c \times V = 0.150 \times 0.0250 = 3.75 \times 10^{-3}\ \text{mol}$
3. Each $CaCl_2$ gives two Cl^- ions: $n(Cl^-) = 2 \times 3.75 \times 10^{-3} = 7.50 \times 10^{-3}\ \text{mol}$
4. $N = 7.50 \times 10^{-3} \times 6.02 \times 10^{23} = 4.52 \times 10^{21}$ ions

Making a standard solution

A standard solution is one whose concentration is known accurately. The aim is to get every bit of the weighed solid into a known final volume.

- **Weigh:** weigh a weighing bottle containing the solid, tip the solid into a beaker, and reweigh the bottle. The difference is the mass transferred. (Alternatively, rinse the weighing bottle into the beaker with distilled water.)
- **Dissolve:** add about 100 cm^3 of distilled water and stir with a glass rod until all the solid has dissolved. Warm gently if needed, and let the solution cool before the next step, because volume changes with temperature.
- **Transfer:** pour the solution through a funnel into a volumetric flask (for example 250 cm^3). Rinse the beaker, rod and funnel with distilled water and add the rinsings to the flask, so that no solute is left behind.
- **Make up to the mark:** add distilled water until the bottom of the meniscus sits on the graduation line, using a dropping pipette for the last few drops, with your eye level with the mark.
- **Mix:** stopper the flask and invert it several times so that the concentration is the same throughout.

Dilution

Adding water to a solution does not change the amount of solute, it only spreads it through a bigger volume. So $n = c_1V_1 = c_2V_2$. The new volume is the original volume plus the volume of water added. To dilute accurately, measure the original solution with a volumetric pipette (it has a much smaller uncertainty than a measuring cylinder) and make it up in a volumetric flask.

Diluting an acid

1. What volume of water must be added to 10.0 cm^3 of 2.00 mol dm^{-3} hydrochloric acid to dilute it to 0.250 mol dm^{-3}?
2. Amount of HCl, which does not change: $n = 2.00 \times 0.0100 = 0.0200 \text{ mol}$
3. Volume needed at the new concentration: $V = \dfrac{n}{c} = \dfrac{0.0200}{0.250} = 0.0800 \text{ dm}^3 = 80.0 \text{ cm}^3$
4. Water to add $= 80.0 - 10.0 = 70.0 \text{ cm}^3$

The concentration fell by a factor of 8, so the final volume must be 8 times the starting volume.

6. Parts per million

Very low concentrations, such as pollutants in water or trace gases in the air, are given in **parts per million (ppm)**, which avoids awkward small numbers.

$$\text{ppm by mass} = \frac{\text{mass of substance}}{\text{total mass of mixture}} \times 10^6 \qquad \text{ppm by volume} = \frac{\text{volume of gas}}{\text{total volume of gas mixture}} \times 10^6$$

1 ppm by mass is 1 mg in every kg. A dilute aqueous solution has a density of about 1 kg dm^{-3}, so 1 ppm is also roughly 1 mg in each dm^3.

Working with ppm

1. A 250 g sample of fish contains 0.100 mg of mercury. What is its mercury content in ppm by mass?
2. Use the same units for both masses: $0.100 \text{ mg} = 1.00 \times 10^{-4} \text{ g}$
3. $\text{ppm} = \dfrac{1.00 \times 10^{-4}}{250} \times 10^6 = 0.400 \text{ ppm}$

4. The other way round: air contains about 420 ppm of carbon dioxide by volume, so 1.00 dm^3 of air holds $420 \times 10^{-6} \times 1.00 = 4.20 \times 10^{-4}\ dm^3$, which is 0.420 cm^3 of CO_2.

7. Volumes of gases

In a gas the molecules are far apart, so the volume a gas takes up depends almost entirely on how many molecules there are, not on how big they are. This is **Avogadro's law**: equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules. It means one mole of any gas occupies the same volume under the same conditions. At room temperature and pressure (r.t.p.) this **molar volume** is taken as 24.0 $dm^3\ mol^{-1}$.

$$n = \frac{V}{24.0} \quad (V \text{ in } dm^3) \qquad \text{or} \qquad n = \frac{V}{24\,000} \quad (V \text{ in } cm^3)$$

Reacting volumes of gases

Because volume is proportional to amount, the balancing numbers in an equation give the ratio of gas volumes directly, as long as all the volumes are measured at the same temperature and pressure. There is no need to convert to moles.

Reacting gas volumes

1. 40 cm^3 of propane is burned in 250 cm^3 of oxygen. All volumes are measured at r.t.p. What is the total volume of gas at the end? $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$
2. Oxygen used: $5 \times 40 = 200\ cm^3$, so $250 - 200 = 50\ cm^3$ of oxygen is left over.
3. Carbon dioxide formed: $3 \times 40 = 120\ cm^3$
4. At r.t.p. the water is a liquid with a negligible volume, so the gas left is $50 + 120 = 170\ cm^3$.

The ideal gas equation

The molar volume of 24.0 dm^3 only applies at room conditions. For any temperature and pressure use the ideal gas equation, which works for any gas or mixture of gases (then n is the total amount of all the gases):

$$pV = nRT$$

Quantity	Unit you must use	Converting
pressure p	Pa	kPa × 1000 = Pa
volume V	m^3	$cm^3 \times 10^{-6} = m^3$; $dm^3 \times 10^{-3} = m^3$
temperature T	K	°C + 273 = K
gas constant R	8.31 $J\ mol^{-1}\ K^{-1}$	given

COMMON MISTAKE

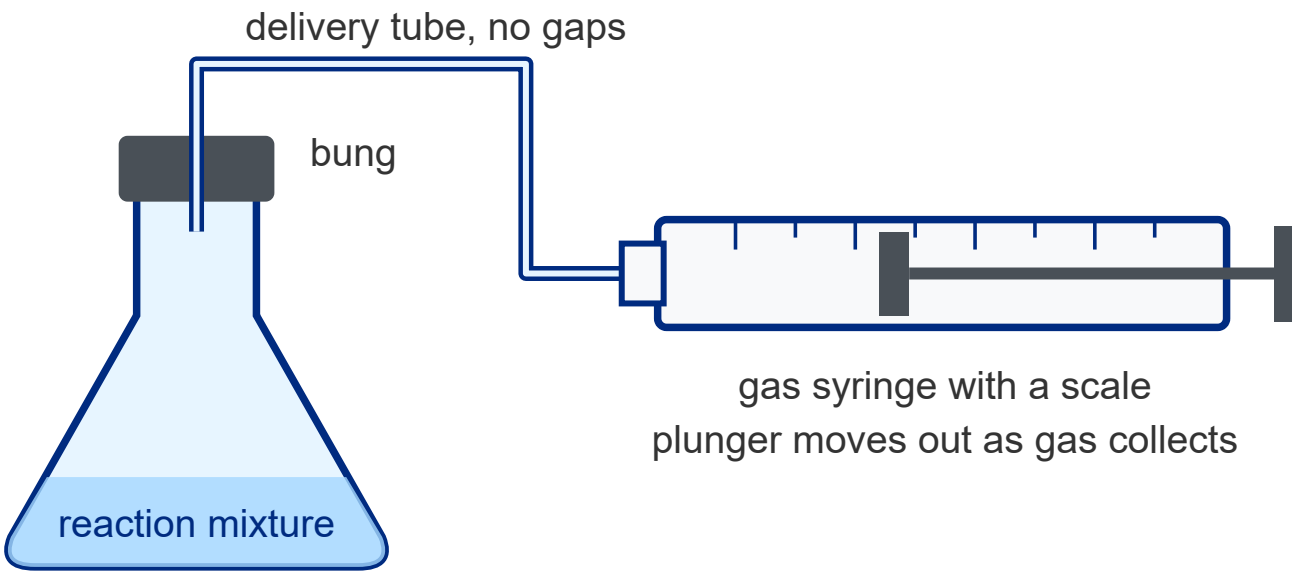
Most wrong answers with $pV = nRT$ come from units. A volume in cm^3 left unconverted is out by a factor of a million, and a temperature in °C gives nonsense. Convert everything to Pa, m^3 and K before you substitute.

Relative molecular mass of a volatile liquid

1. 0.200 g of a volatile liquid is injected into a gas syringe in an oven at 100 °C. It evaporates completely and the vapour occupies 78.0 cm^3 at 101 kPa. Find M_r.
2. Convert the units: $p = 101\,000$ Pa, $V = 78.0 \times 10^{-6}$ m^3, $T = 100 + 273 = 373$ K
3. $n = \dfrac{pV}{RT} = \dfrac{101\,000 \times 78.0 \times 10^{-6}}{8.31 \times 373} = 2.54 \times 10^{-3}$ mol
4. $M = \dfrac{m}{n} = \dfrac{0.200}{2.54 \times 10^{-3}} = 78.7$ g mol^{-1}, so M_r = 78.7. The liquid could be benzene, C_6H_6 (M_r = 78.0).

For a fixed amount of gas, nR is constant, so $\dfrac{p_1V_1}{T_1} = \dfrac{p_2V_2}{T_2}$. This handles a gas moved to new conditions without finding n first. At constant temperature it reduces to $p_1V_1 = p_2V_2$: if 100 cm^3 of gas at 100 kPa is let into an empty 500 cm^3 flask, its pressure falls to $100 \times 100/500 = 20$ kPa.

Measuring a gas volume



Collecting a gas in a gas syringe. The gas volume is read from the scale once the reaction stops; record the room temperature and pressure as well, because the volume depends on both.

A gas syringe lets you measure the volume of gas given off, which can be turned into an amount with $n = V/24.0$ or $pV = nRT$. The same idea is used to measure the molar volume of a gas: react a weighed piece of magnesium ribbon with excess dilute hydrochloric acid and collect the hydrogen. For example, 0.0360 g of Mg ($0.0360/24.3 = 1.48 \times 10^{-3}$ mol) gives the same amount of H_2. If 35.6 cm^3 is collected, the molar volume is $35.6/1.48 \times 10^{-3} = 24\,000$ cm^3 mol^{-1}.

- **Gas lost before the bung goes in:** the volume measured is too small. Keep the reactants apart (for example the solid in a small tube inside the flask) and tip them together after sealing.

- **A sticking plunger:** the gas is not at room pressure when you read the volume.
- **Soluble gases:** carbon dioxide and sulfur dioxide partly dissolve in the water in the flask, so less gas is collected than was made.

8. Moving between quantities

Every quantity you can measure links to amount in moles by one of four routes: mass through the molar mass, number of particles through the Avogadro constant, gas volume through the molar volume (or $pV = nRT$), and solution volume through the concentration. The amount is the hub that connects them all.

KEY IDEA

$$n = \frac{m}{M} \qquad N = nL \qquad n = \frac{V_{\text{gas}}}{24.0\ \text{dm}^3\,\text{mol}^{-1}} \qquad n = cV$$

9. Calculations from equations

Most questions give you a quantity of one substance and ask about another substance in the same reaction. They all follow three steps:

1. Turn the quantity you are given into an amount in moles (mass, gas volume or solution).
2. Use the mole ratio from the balanced equation to find the amount of the substance you want.
3. Turn that amount into the quantity asked for.

Reacting masses and gas volume

1 25.0 g of calcium carbonate is heated until it has fully decomposed. Find the mass of calcium oxide formed and the volume of carbon dioxide at r.t.p.
$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

2 Step 1: $n(\text{CaCO}_3) = \dfrac{25.0}{100.1} = 0.2498\ \text{mol}$

3 Step 2: the ratio is 1 : 1 : 1, so $n(\text{CaO}) = n(\text{CO}_2) = 0.2498\ \text{mol}$

4 Step 3: $m(\text{CaO}) = 0.2498 \times 56.1 = 14.0\ \text{g}$ and $V(\text{CO}_2) = 0.2498 \times 24.0 = 5.99\ \text{dm}^3$

Check with conservation of mass: CO_2 has mass $0.2498 \times 44.0 = 11.0$ g, and 14.0 + 11.0 = 25.0 g.

Gas volume at other conditions

1 What volume of hydrogen, at 25 °C and 100 kPa, is made when 0.500 g of magnesium reacts with excess hydrochloric acid?
$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

2 $n(\text{Mg}) = \dfrac{0.500}{24.3} = 0.02058\ \text{mol}$, and the ratio Mg : H_2 is 1 : 1.

3 $V = \dfrac{nRT}{p} = \dfrac{0.02058 \times 8.31 \times 298}{100\,000} = 5.10 \times 10^{-4}\ \text{m}^3$

4 $5.10 \times 10^{-4}\ \text{m}^3 \times 10^6 = 510\ \text{cm}^3$

When one reactant runs out first

If you are given amounts of two reactants, one may be in **excess**. The other, the **limiting reactant**, is used up first and decides how much product forms. Work out how much of each is needed for the other, and see which one runs short.

Finding the limiting reactant

1. 2.00 g of magnesium is added to 50.0 cm^3 of 1.00 mol dm^{-3} hydrochloric acid. What volume of hydrogen forms at r.t.p.?
2. $n(\text{Mg}) = 2.00 \div 24.3 = 0.0823$ mol and $n(\text{HCl}) = 1.00 \times 0.0500 = 0.0500$ mol
3. Mg + 2HCl: 0.0500 mol of HCl reacts with only 0.0250 mol of Mg. There is 0.0823 mol of Mg, so magnesium is in excess and the acid is limiting.
4. $n(\text{H}_2) = \frac{1}{2} \times 0.0500 = 0.0250$ mol, so $V = 0.0250 \times 24.0 = 0.600\ \text{dm}^3 = 600\ \text{cm}^3$

   Using the magnesium would predict 1.98 dm^3, which is impossible: the acid has run out long before.

10. Acid–base titrations

A titration finds the concentration of a solution by reacting it with a solution of known concentration. In an acid–alkali titration a measured volume of one solution goes in a conical flask with a few drops of indicator, and the other is added from a burette until the indicator just changes colour. That point is the **end point**, and it shows that the acid and alkali have exactly neutralised each other.

- **Measure the solution in the flask** with a volumetric pipette (usually 25.0 cm^3), rinsed first with that solution. Rinse the burette with the solution it will hold, so any water left inside does not dilute it.
- **Read the burette** at eye level, at the bottom of the meniscus, to the nearest 0.05 cm^3. The titre is the final reading minus the initial reading.
- **Swirl and go slowly:** stand the flask on a white tile so the colour change is easy to see, and add the solution drop by drop near the end point.
- **Repeat:** the first run is a rough titration to find roughly where the end point is. Repeat accurately until you have **concordant** titres, within 0.10 cm^3 of each other, and use the mean of those.

Indicator	Colour in acid	Colour in alkali	End point (alkali added to acid)
phenolphthalein	colourless	pink	colourless to the first permanent pale pink
methyl orange	red	yellow	red to orange

For a strong acid with a strong alkali either indicator works, because the pH jumps by several units with a single drop at the end point. Use phenolphthalein when the acid is weak, such as ethanoic acid, and methyl orange when the alkali is weak, such as aqueous ammonia. Universal indicator is never used: it changes colour gradually, so there is no sharp end point.

Concentration from a titration

1. 25.0 cm^3 of sodium hydroxide solution is titrated with 0.100 mol dm^{-3} sulfuric acid. The titres are 22.10 (rough), 21.65, 21.55 and 21.60 cm^3. Find the concentration of the sodium hydroxide.
   $H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$

2. Mean of the concordant titres: $(21.65 + 21.55 + 21.60) \div 3 = 21.60\ \text{cm}^3$

   Leave out the rough titre. These three lie within 0.10 cm^3 of each other.

3. Amount of acid used: $n = 0.100 \times \dfrac{21.60}{1000} = 2.160 \times 10^{-3}\ \text{mol}$

4. The ratio H_2SO_4 : NaOH is 1 : 2, so $n(\text{NaOH}) = 4.320 \times 10^{-3}\ \text{mol}$

5. $c(\text{NaOH}) = \dfrac{4.320 \times 10^{-3}}{0.0250} = 0.173\ \text{mol dm}^{-3}$

11. Percentage yield and atom economy

The amount of product calculated from the equation is the **theoretical yield**, the most you could possibly get. In practice you get less:

$$\text{percentage yield} = \frac{\text{actual mass of product}}{\text{theoretical mass of product}} \times 100$$

Yields fall short because the reaction may not go to completion (it may be reversible, or stopped too soon), side reactions make other products, and some product is lost when it is transferred between containers, filtered or purified.

Atom economy asks a different question: of all the atoms you put in, what fraction ends up in the product you want? It comes straight from the balanced equation, so it is fixed for a given reaction however carefully you work.

$$\text{percentage atom economy} = \frac{\text{molar mass of the desired product}}{\text{sum of the molar masses of all the reactants}} \times 100$$

Use the balancing numbers when you add up the molar masses. A process with a high atom economy makes little waste and uses raw materials efficiently, so it is more sustainable. Reactions with only one product, such as additions, have 100% atom economy. When there is a by-product, finding a use for it or selling it improves the economics of the process.

Yield and atom economy for making lime

1. Heating 50.0 g of calcium carbonate produced 25.2 g of calcium oxide. Find the percentage yield and the atom economy. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

2. Theoretical yield: $n = 50.0 \div 100.1 = 0.4995\ \text{mol}$, so the most CaO possible is $0.4995 \times 56.1 = 28.02\ \text{g}$.

3. $\text{yield} = \dfrac{25.2}{28.02} \times 100 = 89.9\%$

4. $\text{atom economy} = \dfrac{56.1}{100.1} \times 100 = 56.0\%$

   Even at 100% yield, 44% of the mass of the limestone leaves as carbon dioxide. Yield measures how well the practical work went; atom economy measures how wasteful the reaction is.

Two ways of making ethanol show the difference. Fermentation, $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$, has an atom economy of $(2 \times 46.0)/180.0 \times 100 = 51.1\%$. Hydration of ethene, $C_2H_4 + H_2O \rightarrow C_2H_5OH$, has an atom economy of 100%, because every atom ends up in the ethanol.

Quick checks

QUICK CHECK

How many oxygen atoms are there in 0.250 mol of carbon dioxide?

- 1.51×10^{23}
- 3.01×10^{23}
- 6.02×10^{23}
- 4.52×10^{23}

Each CO_2 molecule has two oxygen atoms, so there are $2 \times 0.250 = 0.500$ mol of O atoms, which is $0.500 \times 6.02 \times 10^{23} = 3.01 \times 10^{23}$. The answer 1.51×10^{23} counts molecules, not oxygen atoms.

QUICK CHECK

A compound of nitrogen and oxygen contains 1.40 g of N and 3.20 g of O. Its M_r is 92.0. What is its molecular formula?

- NO_2
- N_2O
- N_2O_4
- NO

N: $1.40/14.0 = 0.100$; O: $3.20/16.0 = 0.200$. The ratio 1 : 2 gives the empirical formula NO_2, with mass 46.0. $92.0/46.0 = 2$, so the molecular formula is N_2O_4. NO_2 is only the empirical formula.

QUICK CHECK

What is the concentration of chloride ions in 0.0500 mol dm^{-3} magnesium chloride solution?

- 0.0250 mol dm^{-3}
- 0.0500 mol dm^{-3}
- 0.150 mol dm^{-3}
- 0.100 mol dm^{-3}

$MgCl_2 \rightarrow Mg^{2+} + 2Cl^-$, so every mole of the salt gives two moles of chloride ions: $2 \times 0.0500 = 0.100$ mol dm^{-3}. 0.150 counts all the ions (Mg^{2+} as well).

QUICK CHECK

What is the ionic equation for the reaction between sodium carbonate solution and hydrochloric acid?

- $CO_3^{2-}(aq) + 2H^+(aq) \rightarrow H_2O(l) + CO_2(g)$ ✓
- $Na_2CO_3(aq) + 2H^+(aq) \rightarrow 2Na^+(aq) + H_2O(l) + CO_2(g)$
- $2Na^+(aq) + 2Cl^-(aq) \rightarrow 2NaCl(aq)$
- $CO_3^{2-}(aq) + H^+(aq) \rightarrow CO_2(g) + OH^-(aq)$

Sodium carbonate is dissolved, so it is written as ions, and Na^+ and Cl^- are spectators. The second option keeps the dissolved sodium carbonate as a formula and leaves spectator Na^+ in the equation.

QUICK CHECK

What volume does 0.100 mol of an ideal gas occupy at 300 K and 100 kPa? (R = 8.31 J mol^{-1} K^{-1})

- 2.49 m^3
- 0.224 dm^3
- 2.49 dm^3 ✓
- 24.9 dm^3

$V = nRT/p = 0.100 \times 8.31 \times 300/100\,000 = 2.49 \times 10^{-3}\ \text{m}^3 = 2.49\ \text{dm}^3$. Leaving the pressure in kPa gives 2.49 m^3, a thousand times too big.

QUICK CHECK

Methyl orange is added to hydrochloric acid in a conical flask, and sodium hydroxide solution is run in from a burette. What colour change marks the end point?

- yellow to orange
- colourless to pink
- red to yellow
- red to orange ✓

Methyl orange is red in acid. The end point is the first orange colour. If the solution has gone yellow you have overshot and added too much alkali. Colourless to pink is phenolphthalein.

QUICK CHECK

Which reaction has an atom economy of 100% for the product shown in bold?

- $CaCO_3 \rightarrow \mathbf{CaO} + CO_2$
- $C_2H_4 + Br_2 \rightarrow \mathbf{C_2H_4Br_2}$
- $Zn + CuSO_4 \rightarrow ZnSO_4 + \mathbf{Cu}$
- $CH_4 + H_2O \rightarrow CO + \mathbf{3H_2}$

An addition reaction has a single product, so every atom of the reactants ends up in it. All the others make a second substance that is not wanted, so their atom economy is below 100% even if the yield is perfect.

QUICK CHECK

A student heats a hydrated salt but does not heat it to constant mass before calculating x, the number of water molecules in the formula. How does this affect the value of x?

- It has no effect, because the mass of the salt is unchanged.
- x is too low, because some water is still in the solid.
- x is too high, because the anhydrous salt seems too heavy.
- x is too high, because extra water is lost.

Water left in the solid makes the "anhydrous" mass too big and the mass of water lost too small. Fewer moles of water per mole of salt means x comes out too low.

AS · Unit 1 · WCH11 Structure, Bonding and Introduction to Organic Chemistry

Atomic structure and the periodic table

Everything about how an element behaves comes from the arrangement of electrons around its nucleus. This topic starts with the particles inside an atom and the mass spectrometer that weighs them, then uses ionisation energies as the experimental evidence for shells, sub-shells and orbitals. It finishes by explaining the repeating patterns, or periodicity, across the periodic table.

1. Protons, neutrons and electrons

An atom has a tiny, dense nucleus containing protons and neutrons, surrounded by electrons that take up almost all of its volume. The nucleus holds nearly all of the mass, because an electron is about 1840 times lighter than a proton.

Particle	Where it is found	Relative mass	Relative charge
proton	nucleus	1	+1
neutron	nucleus	1	0
electron	orbitals around the nucleus	1/1840	−1

- **Atomic (proton) number, Z:** the number of protons in the nucleus. It defines the element: every atom with 11 protons is sodium.
- **Mass number, A:** the total number of protons and neutrons in the nucleus. The number of neutrons is $A - Z$.
- **Atoms are neutral:** an atom has as many electrons as protons. An ion has gained or lost electrons, never protons.

An atom or ion is written with the mass number at the top left and the atomic number at the bottom left, for example $^{7}_{3}\mathrm{Li}$: 3 protons, 3 electrons and $7 - 3 = 4$ neutrons. For an ion, adjust only the electrons: a positive charge means electrons lost, a negative charge means electrons gained.

Species	Protons	Neutrons	Electrons	Why
$^{23}_{11}\mathrm{Na}^{+}$	11	12	10	lost one electron
$^{37}_{17}\mathrm{Cl}^{-}$	17	20	18	gained one electron
$^{27}_{13}\mathrm{Al}^{3+}$	13	14	10	lost three electrons
$^{32}_{16}\mathrm{S}^{2-}$	16	16	18	gained two electrons
$^{1}\mathrm{H}_{2}{}^{16}\mathrm{O}$ molecule	10	8	10	add up the atoms: 2 × H + O

Isotopes

Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons, so they have different mass numbers. Chlorine has two common isotopes, ^{35}Cl (18 neutrons) and ^{37}Cl (20 neutrons).

Isotopes of an element have the **same chemical properties**, because chemical reactions involve electrons and isotopes have the same electron configuration. Their **physical properties** that depend on mass, such as density, melting temperature and the rate at which a gas diffuses, differ slightly.

Relative masses

Single atoms are so light that their masses are compared with a standard, the carbon-12 atom, which is given a mass of exactly 12 units.

- **Relative isotopic mass:** the mass of an atom of one particular isotope compared with one twelfth of the mass of an atom of carbon-12.

- **Relative atomic mass, A_r:** the weighted mean mass of an atom of an element, taking account of the abundance of each isotope, compared with one twelfth of the mass of an atom of carbon-12. This is why A_r is rarely a whole number.
- **Relative molecular mass, M_r:** the mean mass of a molecule compared with one twelfth of the mass of an atom of carbon-12. (For an ionic compound, the relative formula mass.)

2. Mass spectrometry

A mass spectrometer turns the particles in a sample into positive ions, separates the ions according to their **mass-to-charge ratio, m/z**, and measures how many of each arrive at a detector. The result is a mass spectrum: a vertical line for each kind of ion, at its m/z value, with a height showing its **abundance**. Most ions have a single positive charge, so m/z is simply the mass of the ion, which is the mass of the isotope (the electron removed has negligible mass).

A spectrum of an element therefore shows every isotope and its abundance. To label a peak, give the isotope and the charge, for example $^{25}Mg^{+}$. Occasionally two electrons are knocked off; a 2+ ion has half the m/z, so $^{24}Mg^{2+}$ appears at m/z = 12.

Calculating relative atomic mass

$$A_r = \frac{\sum(\text{isotopic mass} \times \text{percentage abundance})}{100} \quad \text{or} \quad A_r = \frac{\sum(\text{isotopic mass} \times \text{relative abundance})}{\sum \text{relative abundances}}$$

Use the first form when the abundances add up to 100%, and the second when you are given peak heights that do not.

Relative atomic mass from percentages

1. A sample of silicon contains 92.2% ^{28}Si, 4.7% ^{29}Si and 3.1% ^{30}Si. Calculate A_r of silicon.
2. Multiply each isotopic mass by its abundance and add: $(28 \times 92.2) + (29 \times 4.7) + (30 \times 3.1) = 2581.6 + 136.3 + 93.0 = 2810.9$
3. Divide by 100: $A_r = 28.1$

   The answer must lie between the lightest and heaviest isotope, and close to the most abundant one. That is a quick check on your arithmetic.

Relative atomic mass from peak heights

1. The mass spectrum of boron has peaks at m/z = 10 and 11, with heights of 23 and 92. Calculate A_r.
2. $\sum(\text{mass} \times \text{height}) = (10 \times 23) + (11 \times 92) = 230 + 1012 = 1242$
3. The heights add up to $23 + 92 = 115$, not 100, so divide by 115: $A_r = 1242 \div 115 = 10.8$

Working back to the abundances

1. Chlorine has isotopes of mass 35 and 37 and A_r = 35.5. Find the percentage of each isotope.
2. Let the percentage of ^{35}Cl be x, so ^{37}Cl is $100 - x$: $\frac{35x + 37(100 - x)}{100} = 35.5$
3. $3700 - 2x = 3550$, so $x = 75$. Chlorine is 75% ^{35}Cl and 25% ^{37}Cl, a ratio of 3 : 1.

Spectra of diatomic molecules

Chlorine is made of Cl_2 molecules, and each of the two atoms in a molecule can be either isotope. That gives three possible molecular ions: ^{35}Cl–$^{35}Cl^+$ at m/z 70, ^{35}Cl–$^{37}Cl^+$ at 72 and ^{37}Cl–$^{37}Cl^+$ at 74. Their relative heights follow from the chance of picking each isotope.

Predicting the Cl_2 peak heights

1. The chance that an atom is ^{35}Cl is $\frac{3}{4}$, and that it is ^{37}Cl is $\frac{1}{4}$.
2. m/z 70 needs ^{35}Cl twice: $\frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$
3. m/z 72 can happen two ways, 35 then 37 or 37 then 35: $2 \times \frac{3}{4} \times \frac{1}{4} = \frac{6}{16}$

   Forgetting the factor of 2 is the most common mistake here.
4. m/z 74 needs ^{37}Cl twice: $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$. The peaks at 70, 72 and 74 are in the ratio **9 : 6 : 1**.

   The spectrum also shows peaks at m/z 35 and 37, in the ratio 3 : 1, from single Cl^+ ions made when some molecules break apart. Bromine is about 50% ^{79}Br and 50% ^{81}Br, so Br_2 gives peaks at 158, 160 and 162 in the ratio 1 : 2 : 1.

Relative molecular mass from a mass spectrum

When a compound goes through a mass spectrometer, many molecules break into fragments, giving a series of peaks. The ion with the largest m/z is usually the whole molecule with one electron removed, the **molecular ion**, M^+. Its m/z value equals the relative molecular mass. For butane, C_4H_{10}, the molecular ion peak is at m/z 58, even though the tallest peak (at 43) is a fragment.

Uses of mass spectrometers

- **Space research:** mass spectrometers on planetary probes identify the elements present, and their isotope ratios, which can differ from those on Earth.
- **Sport:** blood and urine samples are tested for banned drugs, such as anabolic steroids, and for the substances the body breaks them down into.
- **Pharmaceutical industry:** to identify newly made compounds and as a final check on the identity and purity of a drug.
- **Radioactive dating:** measuring the ratio of isotopes such as ^{14}C to ^{12}C to find the age of once-living material.

3. Ionisation energies

DEFINITION

The **first ionisation energy** of an element is the energy needed to remove one electron from each atom in one mole of gaseous atoms, forming one mole of gaseous 1+ ions. For sodium: $Na(g) \rightarrow Na^+(g) + e^-$.

DEFINITION

The **second ionisation energy** is the energy needed to remove one electron from each ion in one mole of gaseous 1+ ions, forming one mole of gaseous 2+ ions. For titanium: $Ti^+(g) \rightarrow Ti^{2+}(g) + e^-$. Each further ionisation energy follows the same pattern.

EXAM TIP

Write the equation the same way for every element: one gaseous particle on the left, the ion one charge higher and one electron on the right. The (g) state symbols are essential, even for an element such as titanium that is a solid at room temperature, and even when the element never normally forms that ion.

Removing an electron means pulling it away from the attraction of the nucleus, so ionisation energies are always endothermic (positive). How much energy is needed depends on how strongly the nucleus attracts the electron being removed:

- **Nuclear charge:** more protons attract the electrons more strongly.
- **Distance from the nucleus:** an electron in a shell further out is attracted less strongly. Attraction falls off quickly with distance.
- **Shielding:** electrons in inner, full shells repel the outer electrons and cancel part of the nuclear attraction. Electrons in the same shell shield one another very little.

Two finer effects explain the small dips you will meet below: which sub-shell the electron is in (a p electron is slightly higher in energy than an s electron in the same shell) and whether it shares its orbital with another electron (two electrons in one orbital repel each other).

Successive ionisation energies

Each successive ionisation energy is larger than the one before. Once an electron has gone, the same number of protons is pulling on fewer electrons, so the remaining electrons are held more tightly and the ion is smaller. Every electron removed is being pulled away from a more positive ion.

The increase is not smooth. Plotted on a log scale, the ionisation energies of an element fall into groups, with a **big jump** each time the next electron has to come from a shell closer to the nucleus. That electron is much nearer the nucleus and has less shielding. The jumps are direct evidence that electrons are arranged in shells, and the number of electrons before the first big jump is the number of outer-shell electrons, which tells you the group.

Finding the group from successive ionisation energies

1. The first five ionisation energies of an element, in kJ mol^{-1}, are 578, 1817, 2745, 11 577 and 14 842. Which group is it in?
2. Compare each value with the one before: ×3.1, ×1.5, **×4.2**, ×1.3. The biggest jump is between the 3rd and 4th ionisation energies.
3. So three electrons are relatively easy to remove and the fourth comes from an inner shell. The element has three outer electrons and is in **group 13** (it is aluminium).

Look for the ratio, not just the difference: later ionisation energies are always bigger in absolute terms.

First ionisation energies across the periodic table

Plot the first ionisation energy against atomic number and a repeating pattern appears, which is an example of periodicity. Each period starts with a group 1 metal at a low point and ends with a noble gas at a peak.

- **Helium is highest of all:** its electrons are in the first shell, as close to the nucleus as possible, with no inner shells to shield them, and it has one more proton than hydrogen.
- **Across a period the general trend is up:** the number of protons increases, but the electrons are added to the same shell, so shielding hardly changes. The outer electrons feel a stronger attraction and are pulled closer.
- **Down a group it falls:** the outer electron is in a shell further from the nucleus and there are more inner shells shielding it. These effects outweigh the extra protons.
- **From a noble gas to the next group 1 metal it drops sharply** (neon to sodium, for example): sodium's outer electron is alone in a new shell, further out and shielded by two full inner shells.

Two small drops break the upward trend in each period, and both are evidence for sub-shells:

Drop	Explanation
Mg (738) → Al (578), and Be (900) → B (801)	Aluminium's outer electron is in the 3p sub-shell, while magnesium's is in 3s. A 3p electron is higher in energy and is slightly shielded by the 3s electrons, so it is easier to remove.
P (1012) → S (1000), and N (1402) → O (1314)	Phosphorus has three 3p electrons, one in each 3p orbital. Sulfur's fourth 3p electron has to pair up in an orbital that already holds one. The two electrons in the same orbital repel each other, so one of them is easier to remove.

The second ionisation energies of successive elements show the same pattern shifted one element to the left, because a 2+ ion forms from an ion that already has one electron fewer. The group 1 elements then sit at the peaks: removing a second electron from Na^+ means breaking into the full 2p sub-shell. Lithium has the biggest second ionisation energy of all, because its second electron comes from the 1s orbital with no shielding and three protons pulling on it.

4. Shells, sub-shells and orbitals

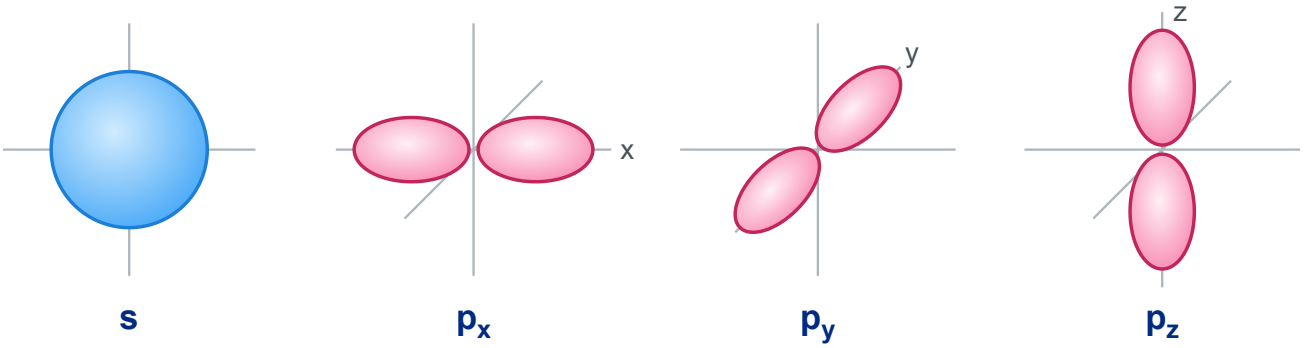
Ionisation energies show that electrons are arranged in **shells** (principal energy levels, numbered n = 1, 2, 3, 4 outwards from the nucleus) and that each shell is split into **sub-shells** labelled s, p, d and f. Each sub-shell is made of **orbitals**.

DEFINITION

An **orbital** is a region around the nucleus where there is a high probability of finding an electron. Each orbital holds at most **two electrons, and they must have opposite spins**.

Sub-shell	Orbitals	Maximum electrons	First appears in shell
s	1	2	n = 1
p	3	6	n = 2
d	5	10	n = 3
f	7	14	n = 4

So the first shell holds 2 electrons (1s), the second 8 (2s, 2p), the third 18 (3s, 3p, 3d) and the fourth 32 (4s, 4p, 4d, 4f). The pattern is $2n^2$.



An s orbital is spherical. A p orbital is dumbbell shaped, and the three p orbitals of a sub-shell lie along the x, y and z axes, at right angles to each other.

Filling the orbitals

- **Lowest energy first:** electrons go into the lowest-energy orbital available. The order is 1s, 2s, 2p, 3s, 3p, **4s, 3d**, 4p. The 4s sub-shell is lower in energy than 3d, so it fills first.

- **Two per orbital, with opposite spins.**
- **Orbitals of equal energy fill singly first:** in a p or d sub-shell, electrons occupy separate orbitals, with parallel spins, before any pairing starts. Electrons repel each other, so they spread out over the available orbitals.

Configurations are written as shell number, sub-shell letter and the number of electrons as a superscript. Oxygen is $1s^2\ 2s^2\ 2p^4$. A shorter form uses the previous noble gas for the inner electrons: sodium, $1s^2\ 2s^2\ 2p^6\ 3s^1$, is [Ne] $3s^1$. In an electrons-in-boxes diagram each box is an orbital and each arrow an electron, with up and down arrows for the two spins.

COMMON MISTAKE

Two elements up to krypton break the simple filling order. Chromium is [Ar] $3d^5\ 4s^1$, not [Ar] $3d^4\ 4s^2$, and copper is [Ar] $3d^{10}\ 4s^1$, not [Ar] $3d^9\ 4s^2$. A half-full or completely full set of d orbitals is a more stable arrangement. Also remember to write 3d before 4s once the 3d sub-shell holds electrons, for example iron is [Ar] $3d^6\ 4s^2$.

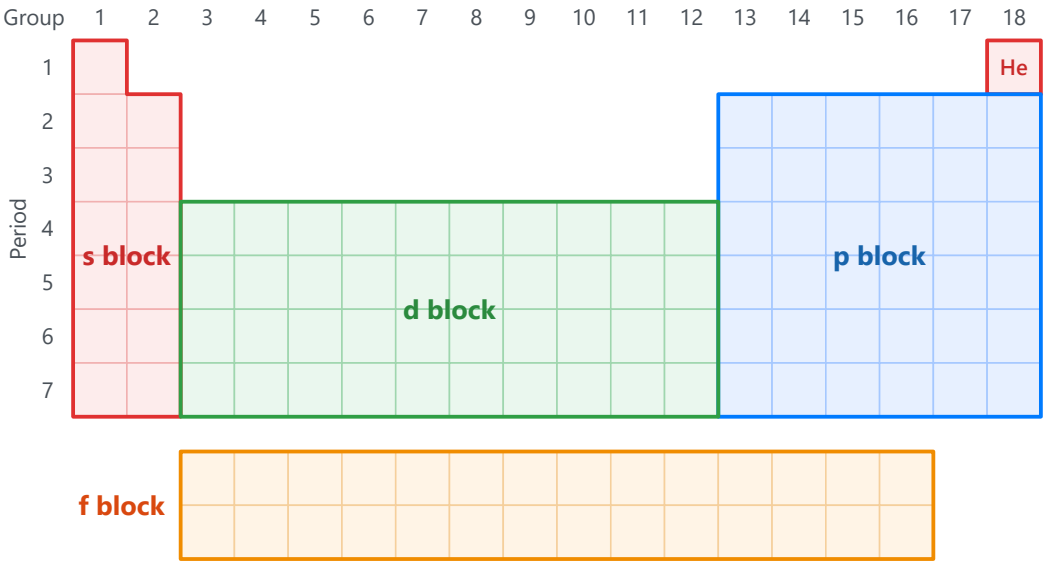
Configurations of ions

For an s- or p-block element, take the atom's configuration and remove electrons from the outermost sub-shell to make a positive ion, or add them to the next available orbitals to make a negative ion. Mg ($1s^2\ 2s^2\ 2p^6\ 3s^2$) becomes Mg^{2+} ($1s^2\ 2s^2\ 2p^6$), and O ($1s^2\ 2s^2\ 2p^4$) becomes O^{2-} ($1s^2\ 2s^2\ 2p^6$). These two ions have the same configuration as neon: they are **isoelectronic**. Further down, Br^- is [Ar] $3d^{10}\ 4s^2\ 4p^6$, the same as krypton, and Ga^{3+} is [Ar] $3d^{10}$, because gallium loses its 4p and both 4s electrons.

5. The periodic table and its blocks

Elements are arranged in order of increasing atomic number. A **period** is a row, and its number is the outer shell being filled. A **group** is a column of elements with the same outer-shell configuration, which is why they react in similar ways: chemistry is all about the outer electrons. Sodium and potassium, for example, both have one outer s electron that they lose easily, so both react vigorously with water to form 1+ ions.

The table divides into **blocks** named after the sub-shell that holds the highest-energy electron. Groups 1 and 2 form the s block, groups 13 to 18 the p block (except helium, $1s^2$, which is placed with the noble gases), the transition elements the d block, and the lanthanides and actinides the f block.



The blocks of the periodic table. The block tells you which type of sub-shell the outermost electrons are filling. Helium ($1s^2$) sits in group 18 but is shaded as an s-block element.

6. Periodicity in periods 2 and 3

Atomic radius

Atomic radius **decreases** across a period, from about 0.16 nm for sodium to 0.10 nm for chlorine. Each element has one more proton than the one before, but the extra electron goes into the same shell, so shielding stays about the same. The greater nuclear charge pulls all the outer electrons closer. The same trend appears in period 2. Down a group, radius increases because each element has an extra shell.

First ionisation energy

For the reasons above, first ionisation energy rises across period 3 from Na (496 kJ mol^{-1}) to Ar (1521 kJ mol^{-1}), with small drops at Mg → Al and P → S. Period 2 shows the same shape, with drops at Be → B and N → O; when you explain those, change 3s and 3p to 2s and 2p.

Melting and boiling temperatures

Melting and boiling temperatures depend on what has to be overcome, which depends on the structure and bonding of each element.

Element	Na	Mg	Al	Si	P (P_4)	S (S_8)	Cl (Cl_2)	Ar
Melting temperature / K	371	923	933	1687	317	388	172	84
Boiling temperature / K	1156	1363	2792	3538	553	718	239	87
Structure	giant metallic			giant covalent	simple molecular			single atoms

- **Na, Mg, Al (giant metallic):** metallic bonding gets stronger from Na to Al. Each atom gives 1, 2 and then 3 electrons to the delocalised sea, and the ions become smaller and more highly charged, so the attraction between the ions and the delocalised electrons increases. The boiling temperatures show this most clearly.
- **Si (giant covalent):** every atom is held by four strong covalent bonds in a giant lattice. Melting means breaking many of these bonds, which takes a great deal of energy, so silicon has the highest melting temperature in the period.
- **P_4, S_8, Cl_2 (simple molecular):** only the weak London forces between molecules are overcome, not the covalent bonds inside them, so the temperatures are low. S_8 has more electrons (128) than P_4 (60), so it has stronger London forces and a higher melting temperature; Cl_2 (34 electrons) is lower still.
- **Ar (single atoms):** there are only very weak London forces between separate atoms with 18 electrons each, so argon has the lowest values.

Period 2 follows the same pattern: Li and Be are metallic, B and C (diamond or graphite) are giant covalent with very high melting temperatures, N_2, O_2 and F_2 are simple molecules that are gases at room temperature, and neon exists as single atoms with the lowest melting temperature of all.

Quick checks

QUICK CHECK

How many protons, neutrons and electrons are in the ion $^{34}_{16}S^{2-}$?

- 16, 18, 16
- 16, 18, 14
- **16, 18, 18**
- 18, 16, 18

Z = 16 gives 16 protons, and $34 - 16 = 18$ neutrons. The 2− charge means two electrons have been gained: $16 + 2 = 18$ electrons. The first option forgets the charge; 14 electrons would be a 2+ ion.

QUICK CHECK

An element has two isotopes, of mass 63 (69.2%) and 65 (30.8%). What is its relative atomic mass?

- 64.0
- **63.6**
- 63.0
- 64.4

$A_r = (63 \times 69.2 + 65 \times 30.8)/100 = 63.6$. The simple average, 64.0, ignores the fact that the lighter isotope is more than twice as abundant.

QUICK CHECK

Chlorine is 75% ^{35}Cl and 25% ^{37}Cl. What are the relative heights of the Cl_2^+ peaks at m/z 70, 72 and 74?

- 3 : 1 : 0
- 9 : 3 : 1
- 1 : 2 : 1
- **9 : 6 : 1**

$\frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$, $2 \times \frac{3}{4} \times \frac{1}{4} = \frac{6}{16}$ and $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$. The 72 peak can form in two ways, which is why it is 6, not 3. A 1 : 2 : 1 ratio belongs to Br_2, whose isotopes are equally abundant.

QUICK CHECK

The first four ionisation energies of an element are 738, 1451, 7733 and 10 543 kJ mol^{-1}. Which group is the element in?

- 2
- 3
- 4
- 1

The big jump comes between the 2nd and 3rd values (1451 to 7733, more than five times bigger). Two electrons come from the outer shell and the third from a full inner shell, so the element is in group 2 (it is magnesium).

QUICK CHECK

Why is the first ionisation energy of oxygen lower than that of nitrogen?

- Oxygen has fewer protons than nitrogen.
- Oxygen's outer electron is in a shell further from the nucleus.
- Oxygen's electron is removed from a 2p orbital that holds two electrons, which repel each other.
- Oxygen's outer electron is in the 2s sub-shell.

Nitrogen has one electron in each 2p orbital. Oxygen's fourth 2p electron has to pair up, and repulsion between the two electrons in that orbital makes one of them easier to remove. Both elements lose their electron from the second shell, and oxygen has more protons, not fewer.

QUICK CHECK

What is the electron configuration of a chromium atom (Z = 24)?

- [Ar] $3d^4\ 4s^2$
- [Ar] $3d^5\ 4s^1$
- [Ar] $3d^6$
- [Ar] $4s^2\ 4p^4$

Chromium is an exception: a half-full 3d sub-shell, with one electron in each of the five d orbitals and one in 4s, is more stable than $3d^4\ 4s^2$, the configuration the simple filling order predicts.

QUICK CHECK

Which period 3 element has the highest melting temperature?

- Aluminium, because it has the strongest metallic bonding
- Sulfur, because S_8 molecules have the most electrons
- Magnesium, because its ions have a 2+ charge
- Silicon, because melting breaks many strong covalent bonds in a giant lattice

Silicon has a giant covalent structure, so melting means breaking strong covalent bonds throughout the lattice. Sulfur is simple molecular: melting only overcomes London forces between S_8 molecules, which are much weaker than covalent bonds.

QUICK CHECK

Why does the first ionisation energy decrease down group 1, even though the nuclear charge increases?

- The outer electron is paired, so it is repelled more.
- The outer electron is further from the nucleus and is shielded by more inner shells.
- The outer electron moves into a p sub-shell.
- Larger atoms have fewer protons per neutron.

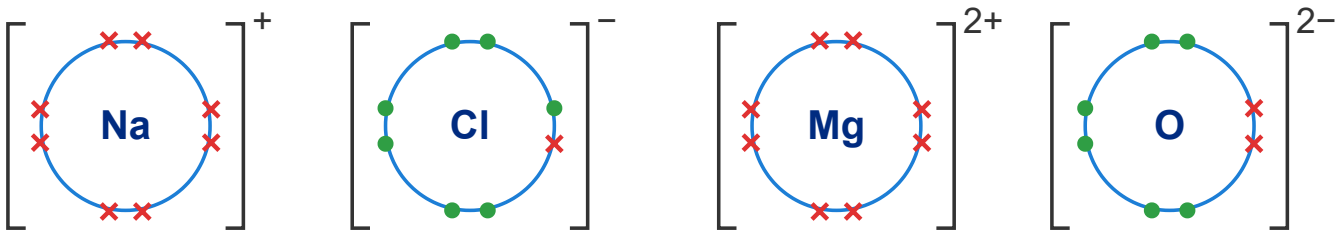
Each step down adds a shell. The outer s electron is further away and has more full shells between it and the nucleus, and these two effects outweigh the extra protons, so less energy is needed to remove it.

Bonding and structure

Every chemical bond is an electrostatic attraction between positive and negative charges. What differs is where the charges are: between ions, between nuclei and shared electrons, or between metal ions and a sea of electrons. This topic explains each type of bond, the shapes that molecules take, how bonds become polar, and how the structure of a substance explains its melting temperature, conductivity and solubility.

1. Ionic bonding

When a metal reacts with a non-metal, electrons are transferred. The metal atoms lose their outer electrons to become positive ions, and the non-metal atoms gain electrons to become negative ions. In simple cases each ion ends up with the electron configuration of a noble gas: sodium (2,8,1) becomes Na^+ (2,8) and chlorine (2,8,7) becomes Cl^- (2,8,8). In terms of sub-shells, Mg $1s^2\ 2s^2\ 2p^6\ 3s^2$ becomes Mg^{2+} $1s^2\ 2s^2\ 2p^6$, and O $1s^2\ 2s^2\ 2p^4$ becomes O^{2-} $1s^2\ 2s^2\ 2p^6$.



sodium chloride, NaCl magnesium oxide, MgO

Dot-and-cross diagrams for two ionic compounds, showing outer shells only. Crosses are electrons that came from the metal, dots those that came from the non-metal. Each ion has a full outer shell, and the brackets and charges show that the ions are separate.

DEFINITION

Ionic bonding is the strong electrostatic attraction between oppositely charged ions.

The attraction acts equally in every direction, so each ion attracts as many oppositely charged ions as can pack around it. The result is a **giant ionic lattice**, a regular repeating arrangement of a huge number of ions. In sodium chloride every Na^+ is surrounded by six Cl^- ions and every Cl^- by six Na^+ ions. The formula NaCl gives only the ratio of ions; there are no molecules.

What makes ionic bonding stronger

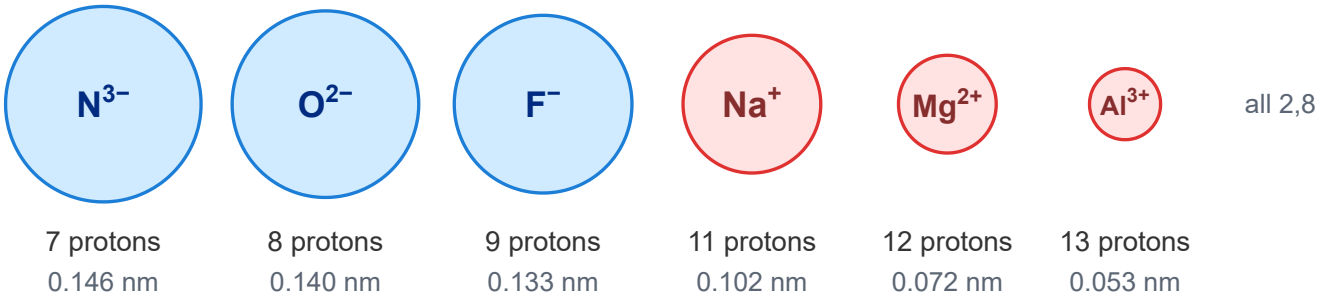
The attraction between two ions is greater when their charges are larger and when their centres are closer together, which happens when the ions are smaller. Stronger attraction means more energy is needed to separate the ions, so the melting temperature is higher.

Compound	Ions	Melting temperature / °C	Why

NaF	Na^+, F^-	993	Same charges; the halide ion gets bigger from F^- to I^-, so the ions are further apart and the attraction weaker.
NaCl	Na^+, Cl^-	801	
NaBr	Na^+, Br^-	747	
NaI	Na^+, I^-	661	
MgO	Mg^{2+}, O^{2-}	2852	Both ions are doubly charged, and Mg^{2+} is smaller than Na^+, so the attraction is much stronger than in NaCl.

Ionic radii

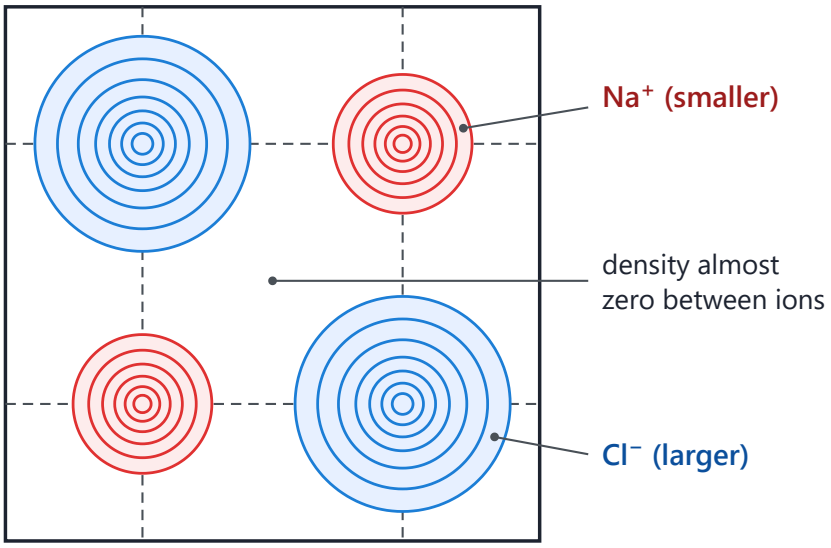
- **Positive ions are smaller than their atoms.** The atom loses its whole outer shell, and the same number of protons now pulls on fewer electrons, so they are held closer.
- **Negative ions are larger than their atoms.** The extra electrons repel the others, and the same nuclear charge is shared among more electrons, so the attraction per electron is less.
- **Down a group, ionic radius increases**, because each ion has one more shell of electrons.
- **Isoelectronic ions** (with the same number of electrons) get smaller as the number of protons increases.



These ions all have the electron configuration $1s^2\ 2s^2\ 2p^6$, the same as neon. From N^{3-} to Al^{3+} the number of protons rises from 7 to 13, so the same ten electrons are pulled in more tightly and the ions get smaller. Circles are drawn to scale.

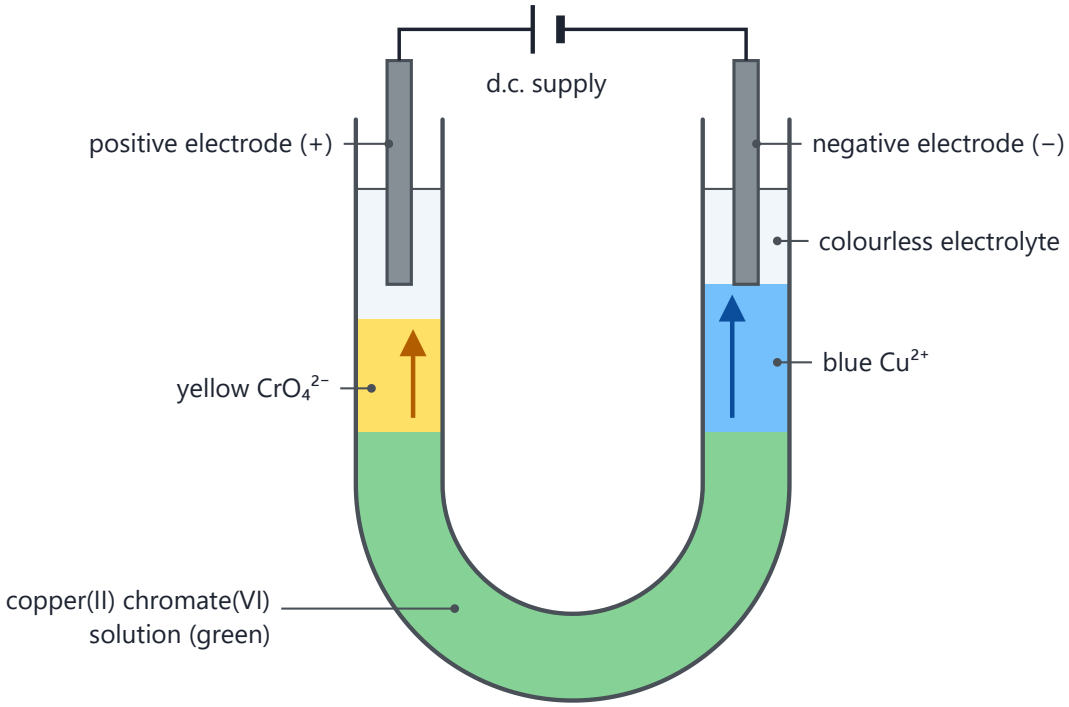
Evidence that ions exist

Electron density maps. When X-rays pass through a crystal they are scattered by the electrons, and the pattern can be turned into a map of electron density, drawn with contour lines like a map of hills. For sodium chloride the map shows separate, almost spherical regions of electron density, arranged regularly, with the density falling almost to zero between neighbouring ions. That is what separate ions would look like. The chloride ions appear larger than the sodium ions. The map does not show a sharp edge to each ion, so ionic radii cannot be read exactly from it.



An electron density map of sodium chloride. Each set of rings is one ion; the larger ones are Cl^- and the smaller ones Na^+.

Migration of ions. If ions exist, a potential difference should move them: positive ions towards the negative electrode, negative ions towards the positive one. Green copper(II) chromate(VI) solution placed under a colourless electrolyte in a U-tube shows this. After some time a blue colour (Cu^{2+} ions) rises towards the negative electrode and a yellow colour (CrO_4^{2-} ions) towards the positive electrode. Similarly, a crystal of potassium manganate(VII) on filter paper moistened with an electrolyte, connected to a low-voltage d.c. supply, gives a purple streak that moves towards the positive electrode, because the coloured particle is the negative MnO_4^- ion.



Coloured ions move in opposite directions: blue Cu^{2+} to the negative electrode and yellow CrO_4^{2-} to the positive electrode.

Properties of ionic compounds

- **High melting and boiling temperatures:** melting means overcoming strong electrostatic attractions between many ions throughout the giant lattice, which needs a lot of energy.
- **Conduct electricity when molten or dissolved, but not when solid:** the ions carry the charge. In the solid they are held in fixed positions; when the lattice breaks down on melting or dissolving, they are free to move.
- **Often soluble in water:** polar water molecules are attracted to the ions and pull them out of the lattice (the details come later in the course).
- **Hard but brittle:** a sharp blow can shift one layer of ions relative to the next. Ions of the same charge then line up next to each other, they repel, and the crystal splits apart.

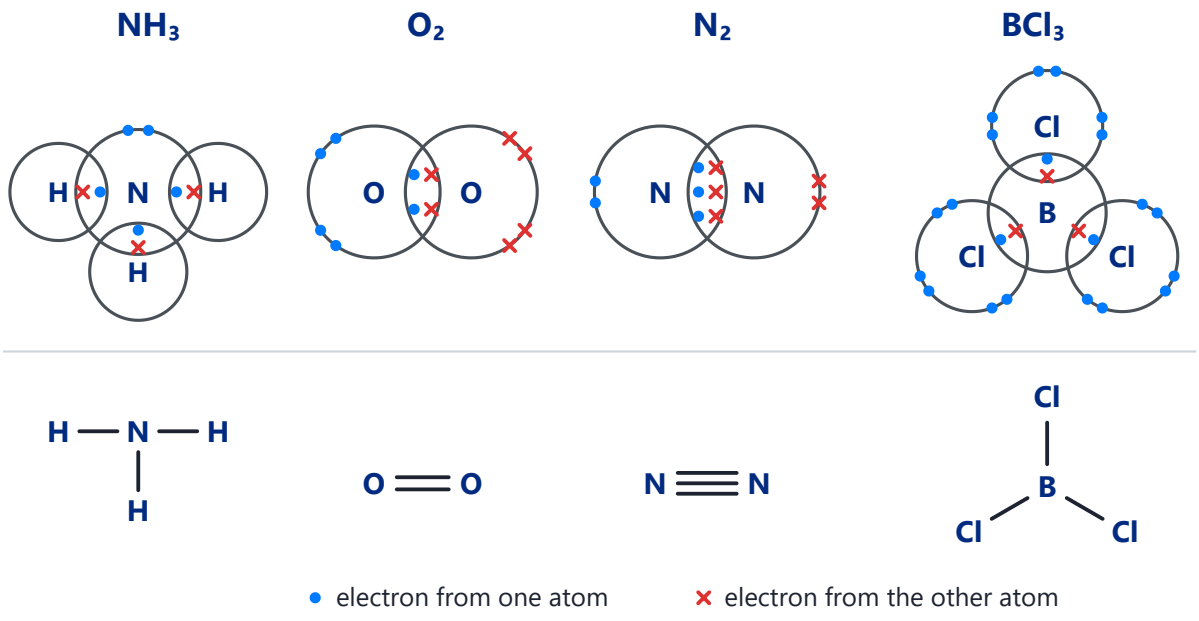
2. Covalent bonding

DEFINITION

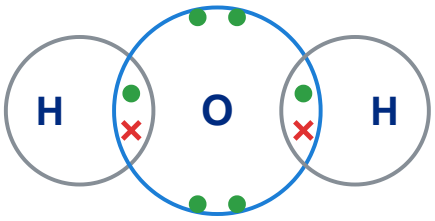
A **covalent bond** is the strong electrostatic attraction between a shared pair of electrons and the nuclei of the two bonded atoms.

Covalent bonds form between non-metal atoms. Electron density maps of molecules such as H_2 show a high concentration of electron density between the two nuclei. That negative charge is attracted to both positive nuclei, holding them together, and these attractions outweigh the repulsion between the two nuclei. Each atom usually gains a share of enough electrons to fill its outer shell.

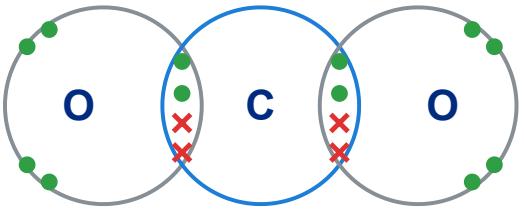
Atoms can share more than one pair. Oxygen, O=O, has a double bond (two shared pairs), and nitrogen, N≡N, has a triple bond (three shared pairs). The bond is strong: it takes a lot of energy to break, which is why giant covalent structures such as diamond have such high melting temperatures. The forces *between* separate molecules are a different and much weaker thing.



Dot-and-cross diagrams of NH_3, O_2, N_2 and BCl_3 with their displayed formulae. Ammonia has three bonding pairs and one lone pair; oxygen shares two pairs and nitrogen three; boron in BCl_3 has only six outer electrons.



water, H_2O



carbon dioxide, O=C=O

Water has two O–H bonding pairs and two lone pairs on oxygen. In carbon dioxide each oxygen shares two pairs with carbon, making two double bonds, and keeps two lone pairs. Dots are electrons from O, crosses from H or C.

Not every atom ends up with eight outer electrons. Boron in BF_3 and BCl_3 has only six, which is why these molecules readily accept a lone pair. Elements in period 3 and below can share more than eight: phosphorus in PCl_5 has ten outer electrons and sulfur in SF_6 has twelve.

Bond length and bond strength

The **bond length** is the distance between the two nuclei. At that distance the attraction of both nuclei for the shared electrons exactly balances the repulsion between the two nuclei (and between the inner electrons). A double or triple bond puts more electron density between the nuclei, so the attraction is greater: the nuclei are pulled closer together and the bond is harder to break.

Bond	Shared pairs	Bond length / nm	Mean bond enthalpy / kJ mol^{-1}
C–C	1	0.154	347
C=C	2	0.134	612
C≡C	3	0.120	838

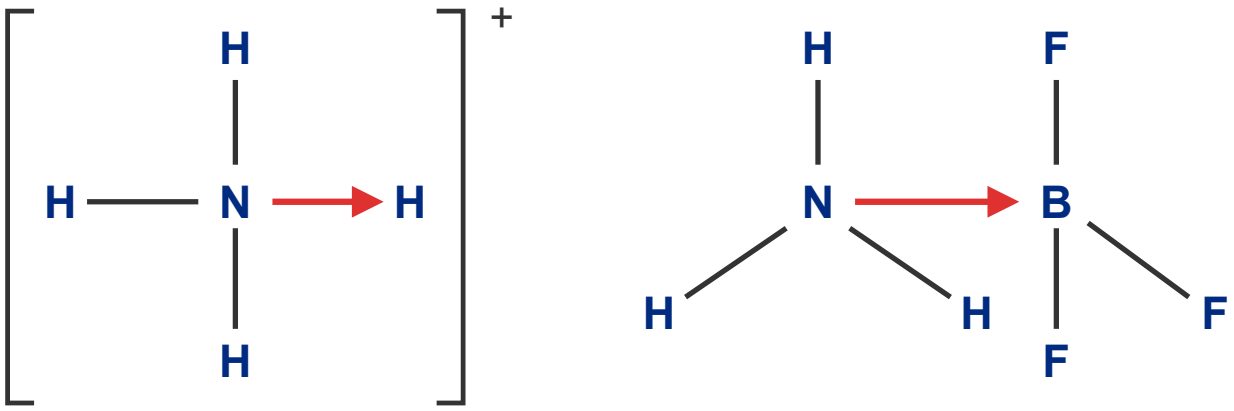
KEY IDEA

More shared pairs between two atoms means more electron density between the nuclei, a stronger attraction, a **shorter** bond and a **stronger** bond. Note that a double bond is not twice as strong as a single bond.

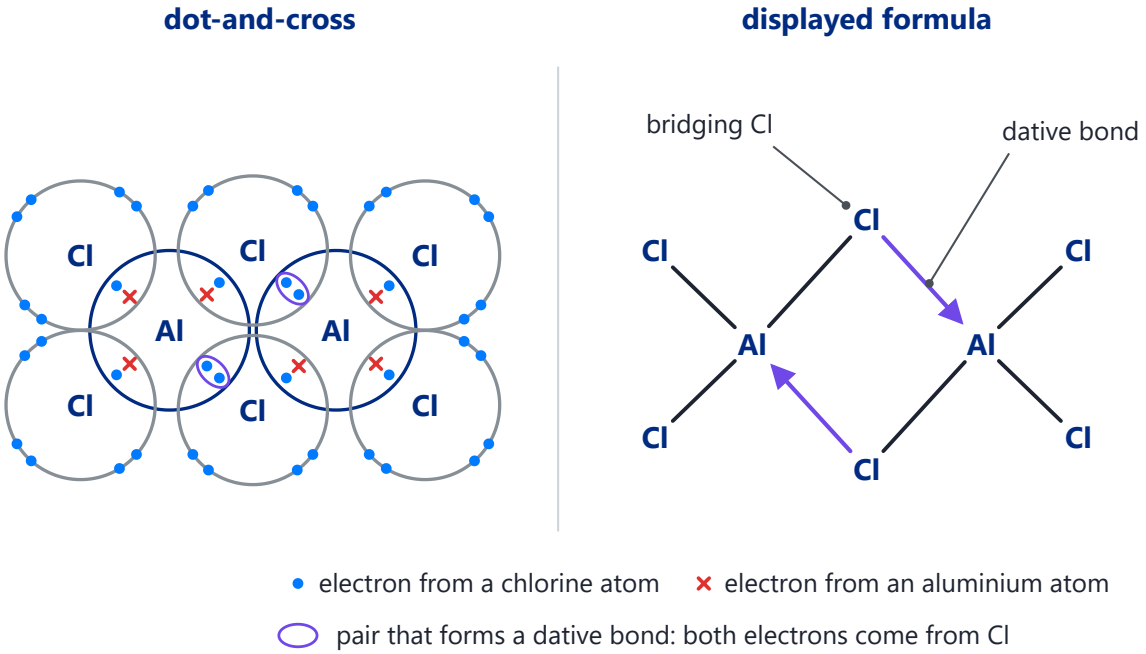
Dative covalent (coordinate) bonds

In a **dative covalent bond** both electrons of the shared pair come from the same atom. It needs one atom with a lone pair and another with an empty space in its outer shell. In a displayed formula it is drawn as an arrow pointing from the atom that donates the lone pair to the atom that accepts it. Once formed, a dative bond is identical to any other covalent bond.

- **Ammonium ion, NH_4^+:** $NH_3 + H^+ \rightarrow NH_4^+$. The nitrogen's lone pair bonds to an H^+ ion, which has no electrons. All four N–H bonds are then the same, and the ion is tetrahedral.
- **Oxonium ion, H_3O^+:** $H_2O + H^+ \rightarrow H_3O^+$. One of oxygen's two lone pairs forms the bond, so one lone pair is left.
- **NH_3BF_3:** the lone pair on the nitrogen of NH_3 fills the gap in the outer shell of boron in BF_3, written $H_3N \rightarrow BF_3$.
- **Aluminium chloride, Al_2Cl_6:** at lower temperatures, two $AlCl_3$ molecules join. A lone pair on a chlorine of each molecule forms a dative bond to the aluminium of the other, giving each aluminium four bonding pairs.



Dative bonds (red arrows) in the ammonium ion and in $H_3N{\rightarrow}BF_3$. Each arrow points away from the nitrogen, which supplies both electrons of the bond.



The Al_2Cl_6 dimer: two bridging chlorine atoms each use a lone pair to form a dative bond (arrow) to the other aluminium.

3. Shapes of molecules and ions

The shape of a molecule is set by the electron pairs in the outer shell of its central atom. Pairs of electrons are regions of negative charge, so they repel one another and move as far apart as possible. This is **electron-pair repulsion theory**. Count the pairs and you know their arrangement; then the positions of the atoms give the shape.

- **Lone pairs repel more strongly than bonding pairs.** A lone pair is held by only one nucleus, so it sits closer to the central atom and spreads out more. The order of repulsion is lone pair–lone pair > lone pair–bonding pair > bonding pair–bonding pair.
- **Each lone pair squeezes the bond angles** by roughly 2.5°: 109.5° in CH_4, 107° in NH_3 and 104.5° in H_2O.
- **A double or triple bond counts as one bonding region**, because its electrons all point in the same direction. CO_2 has two bonding regions around carbon, so it is linear; each carbon in ethene, C_2H_4, has three, so the H–C–H angles are 120°.
- **The shape is named from the atoms only,** even though the lone pairs help to decide it. Ammonia has four pairs arranged tetrahedrally, but its shape is pyramidal.

Bonding pairs	Lone pairs	Shape	Bond angle	Examples
2	0	linear	180°	$BeCl_2$, CO_2, HCN
3	0	trigonal planar	120°	BF_3, $AlCl_3$, SO_3, NO_3^-, CO_3^{2-}
4	0	tetrahedral	109.5°	CH_4, $SiCl_4$, NH_4^+, SO_4^{2-}
3	1	trigonal pyramidal	107°	NH_3, PCl_3, H_3O^+
2	2	bent (V-shaped)	104.5°	H_2O, OF_2
5	0	trigonal bipyramidal	120° and 90°	PCl_5 (gas), PF_5
6	0	octahedral	90°	SF_6
4	2	square planar	90°	XeF_4

With five or six pairs, lone pairs take the positions that keep them furthest from other pairs. That gives a few more shapes you can work out without needing to learn their names: ClF_3 (3 bonding, 2 lone) is T-shaped, SF_4 (4 bonding, 1 lone) is a distorted tetrahedron, XeF_2 and I_3^- (2 bonding, 3 lone) are linear, and BrF_5 (5 bonding, 1 lone) is square pyramidal.

Explaining the shape of ammonia

1. Nitrogen has five outer electrons. Three are shared with the three hydrogen atoms, which leaves one lone pair. So there are **3 bonding pairs and 1 lone pair**.
2. The four electron pairs repel each other and move as far apart as possible, into a tetrahedral arrangement.
3. The lone pair repels more strongly than the bonding pairs, so it pushes the N–H bonds slightly closer together.
4. The shape is **trigonal pyramidal** and the H–N–H bond angle is **107°**, smaller than the 109.5° of a regular tetrahedron.

Exam answers need all of these: the number of each kind of pair, that pairs repel to get as far apart as possible, that lone pairs repel more (or, with no lone pairs, that the pairs repel equally), and then the name of the shape and the angle.

Predicting the shape of an ion: H_3O^+

1. Start with the central atom's outer electrons: oxygen has 6.
2. Add one electron for each atom bonded by a single bond (3 H), then adjust for the charge: a 1+ charge means one electron fewer. $6 + 3 - 1 = 8$ electrons, which is **4 pairs**.
3. Three pairs bond to H atoms, so there are 3 bonding pairs and 1 lone pair: the same as ammonia.
4. H_3O^+ is trigonal pyramidal, with bond angles of about 107°.

Predicting the shape of XeF_4

1. Xenon has 8 outer electrons, and the four fluorine atoms add one each: $8 + 4 = 12$ electrons, so **6 pairs**.
2. Four pairs bond to F atoms, leaving 2 lone pairs. Six pairs point to the corners of an octahedron.
3. The two lone pairs repel each other most, so they take opposite positions, 180° apart, above and below the plane.
4. The four F atoms lie in a flat square around Xe: the shape is **square planar** and the F–Xe–F bond angle is **90°**.

4. Electronegativity and polarity

DEFINITION

Electronegativity is the ability of an atom to attract the bonding pair of electrons in a covalent bond towards itself.

Electronegativity is measured on the Pauling scale. Fluorine is the most electronegative element, with a value of 4.0, followed by oxygen, nitrogen and chlorine.

- **It increases across a period:** the nuclear charge increases while the bonding electrons stay in the same shell with similar shielding, and the atoms get smaller, so the bonding pair is pulled in more strongly.
- **It decreases down a group:** the bonding pair is further from the nucleus and more shielded by inner shells.

Element	H	C	N	O	F	Na	Mg	Al	Si	P	S	Cl	Br	I
Pauling value	2.1	2.5	3.0	3.5	4.0	0.9	1.2	1.5	1.8	2.1	2.5	3.0	2.8	2.5

Polar bonds and the bonding continuum

When two identical atoms are bonded, as in H_2 or Cl_2, the electrons are shared equally and the bond is **non-polar**. When the atoms have different electronegativities, the bonding pair is pulled towards the more electronegative atom. That atom gains a small negative charge, δ−, and the other is left with a small positive charge, δ+. The bond is a **polar covalent bond** and has a permanent dipole: in hydrogen chloride, $H^{\delta+}$–$Cl^{\delta-}$.

Pure covalent and pure ionic bonding are the two ends of a continuous scale. The bigger the difference in electronegativity, the more uneven the sharing. A small difference gives a slightly polar bond, and a very large difference (more than about 1.7) means the electron is effectively transferred and the bonding is mainly ionic. Most real compounds lie somewhere in between. Ionic compounds can also have some covalent character: a small, highly charged positive ion can distort, or **polarise**, the electron cloud of a large negative ion, pulling electron density into the space between them.

Polar and non-polar molecules

A molecule with polar bonds is not necessarily polar overall. Each bond dipole has a direction, and if the molecule is symmetrical the dipoles cancel out. CO_2 is linear, so its two C=O dipoles point in opposite directions and cancel; BF_3 (trigonal planar) and CCl_4 (tetrahedral) cancel in the same way, so these molecules are non-polar. Water is bent, so its two O–H dipoles add up to give a polar molecule with δ− on the oxygen side. Replacing one Cl in CCl_4 by H gives $CHCl_3$, which is no longer symmetrical and is polar.

KEY IDEA

A molecule is polar when it has polar bonds **and** a shape in which their dipoles do not cancel. Symmetrical molecules with identical bonds and no lone pairs on the central atom are non-polar, even if every bond is polar.

A simple test: run a thin jet of a liquid from a burette past an electrically charged rod. A jet of a polar liquid such as water bends towards the rod, because the molecules turn so that their oppositely charged ends face it. This happens whether the rod is positive or negative. A jet of a non-polar liquid such as cyclohexane or tetrachloromethane is not deflected. The more polar the molecules, the greater the deflection.

5. Metallic bonding

DEFINITION

Metallic bonding is the strong electrostatic attraction between metal cations and a sea of delocalised electrons.

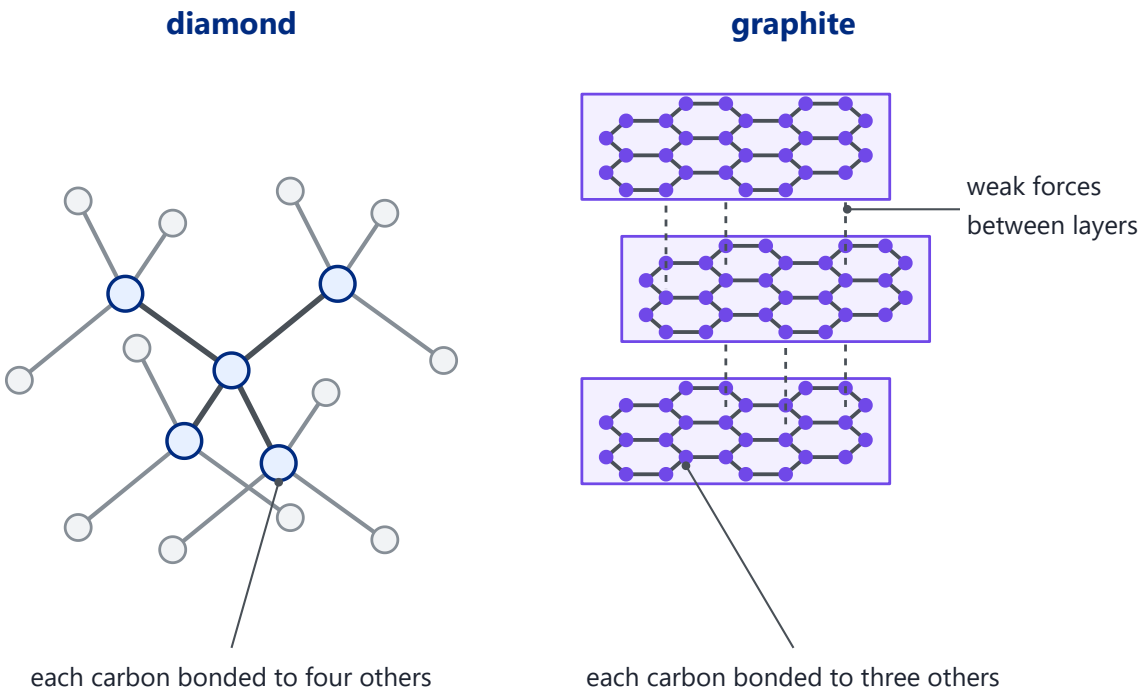
In a solid metal, each atom releases its outer electrons into a shared "sea" that is spread through the whole structure. These electrons are **delocalised**: they do not belong to any one atom. What is left is a regular, closely packed lattice of positive ions held together by their attraction to the delocalised electrons. This is a **giant metallic lattice**.

Metallic bonding is stronger when there are more delocalised electrons per atom, when the ions have a greater charge, and when the ions are smaller. Magnesium has a higher melting temperature (650 °C) than sodium (98 °C): each Mg atom gives two electrons to the sea and forms a smaller 2+ ion, while Na gives one electron and forms a larger 1+ ion.

- **High melting temperatures:** lots of energy is needed to overcome the strong attraction between the ions and the delocalised electrons.
- **Good electrical conductors, as solids and liquids:** the delocalised electrons can move through the structure when a potential difference is applied.
- **Good thermal conductors:** the mobile electrons carry kinetic energy quickly through the metal.
- **Malleable and ductile:** all the ions are the same, so when a force makes layers of ions slide over each other, the surroundings of every ion stay the same and the bonding is not broken. The metal changes shape instead of shattering.

6. Giant covalent structures

In a giant covalent (macromolecular) lattice, every atom is joined to its neighbours by strong covalent bonds in a network that runs through the whole crystal. To melt it, many of these bonds must be broken, so the melting temperatures are very high. These substances are insoluble in water, because water molecules cannot provide the energy needed to break covalent bonds.



Diamond (left): each carbon is bonded to four others in a rigid tetrahedral network. Graphite (right): each carbon is bonded to three others in flat hexagonal layers, which are stacked with weak forces between them.

Structure	Bonding	Properties and why
Diamond	Each C forms four covalent bonds in a tetrahedral network.	Very high melting temperature and extremely hard, because the rigid network of strong bonds runs in all directions. Does not conduct: all four outer electrons of each carbon are held in bonds, so none are free to move.
Graphite	Each C forms three covalent bonds in flat layers of hexagons. The fourth outer electron is delocalised over the layer.	Very high melting temperature (strong bonds within the layers). Conducts along the layers, because the delocalised electrons move freely within a layer, but not well across them. Soft and slippery: the layers are held together only by weak London forces, so they slide over each other easily.
Graphene	A single layer of graphite, one atom thick.	Extremely strong for its mass, because of the network of strong covalent bonds. Conducts electricity very well across the sheet, because one electron per carbon is delocalised. Almost transparent.
Carbon nanotubes	A sheet of graphene rolled into a tube a few nanometres wide.	Very high tensile strength and conduct electricity along the tube, for the same reasons as graphene. Possible uses include strengthening materials, electronics, and carrying drug molecules into cells.
Fullerenes (e.g. C_{60})	Separate ball-shaped molecules of 60 carbon atoms, with delocalised electrons.	A simple molecular substance: only weak London forces act between the molecules, so it turns to vapour at a far lower temperature than diamond or graphite and dissolves in some organic solvents.
Silicon(IV) oxide, SiO_2	Each Si is bonded to four O atoms and each O to two Si atoms in a tetrahedral network.	Very high melting temperature (about 1700 °C), hard, insoluble, and does not conduct, for the same reasons as diamond.

7. Simple molecular structures

Most covalent substances are made of separate small molecules. The bonds inside each molecule are strong, but the forces *between* molecules (London forces, permanent dipole forces and hydrogen bonds, covered in the topic on intermolecular forces) are weak. Melting or boiling only overcomes these intermolecular forces, not the covalent bonds, so simple molecular substances have low melting and boiling temperatures and many are gases or liquids at room temperature. They do not conduct electricity, because the molecules are neutral and no electrons are free to move.

- **Iodine, I_2:** a solid in which I_2 molecules are packed in a regular lattice, held only by London forces. It melts at 114 °C and gives a purple vapour when warmed.
- **Ice, H_2O:** each water molecule forms two covalent O–H bonds and takes part in hydrogen bonds to its neighbours, with each oxygen surrounded tetrahedrally by four hydrogens. This open arrangement holds the molecules further apart than in liquid water, which is why ice is less dense than water and floats.

COMMON MISTAKE

Only use the words "molecules" and "intermolecular forces" for simple molecular substances. Melting sodium chloride overcomes ionic bonds, melting a metal overcomes metallic bonding and melting diamond or silicon(IV) oxide breaks covalent bonds. Writing that "intermolecular forces break" when diamond melts loses the mark.

8. Structure, bonding and properties

Property	Giant ionic	Simple molecular	Giant covalent	Giant metallic
Particles	positive and negative ions	small molecules	atoms	positive ions in a sea of delocalised electrons
What holds them	ionic bonds throughout the lattice	weak intermolecular forces between molecules	strong covalent bonds throughout	metallic bonding throughout
Melting temperature	high	low	very high	usually high
Conducts as a solid?	no: ions fixed	no	no (except graphite and graphene)	yes: delocalised electrons move
Conducts when molten?	yes: ions free to move	no	no	yes
Solubility in water	often soluble	usually insoluble or slightly soluble (some polar molecules dissolve)	insoluble	insoluble (some react)
Examples	NaCl, MgO	I_2, ice, CO_2, CH_4	diamond, graphite, SiO_2, Si	Na, Mg, Cu

Identifying a structure from data

1. A white solid melts at 1710 °C. It does not conduct electricity as a solid or when molten, and it does not dissolve in water. What type of structure does it have?

2. The very high melting temperature rules out a simple molecular structure: something strong holds every particle in place.

3. It does not conduct when molten, so it has no ions that become mobile on melting: it is not ionic. It does not conduct as a solid, so it is not a metal.

4. It must be **giant covalent**. (These are the properties of silicon(IV) oxide.)

Check conductivity in both states: solid only tells metal and graphite apart from the rest; molten tells ionic compounds apart from giant covalent ones.

Quick checks

QUICK CHECK

What is the shape of a PCl_3 molecule, and its approximate bond angle?

- trigonal planar, 120°
- trigonal pyramidal, 107°
- tetrahedral, 109.5°
- T-shaped, 90°

Phosphorus has 5 outer electrons; three bond to Cl, leaving one lone pair. Four pairs give a tetrahedral arrangement, and the lone pair squeezes the bonds to about 107°, so the shape is pyramidal. Trigonal planar would need three pairs and no lone pair, as in BCl_3.

QUICK CHECK

Why does magnesium oxide have a much higher melting temperature than sodium chloride?

- MgO has strong covalent bonds between its atoms.
- MgO molecules have stronger intermolecular forces.
- Mg is more reactive than Na.
- Mg^{2+} and O^{2-} have higher charges and Mg^{2+} is smaller, so the ionic attraction is stronger.

Both compounds are giant ionic lattices. Doubly charged ions, closer together, attract each other much more strongly, so more energy is needed to separate them. MgO has no molecules, so "intermolecular forces" is wrong.

QUICK CHECK

Which of these molecules is polar?

- CO_2
- CCl_4
- $CHCl_3$
- BF_3

All four contain polar bonds, but in CO_2, CCl_4 and BF_3 the symmetrical shape makes the bond dipoles cancel. In $CHCl_3$ one position holds H instead of Cl, so the dipoles no longer cancel and the molecule is polar.

QUICK CHECK

Which ion has the largest radius?

- O^{2-}
- F^-
- Na^+
- Mg^{2+}

All four have 10 electrons. Oxide has the fewest protons (8), so its electrons are held least tightly and it is the biggest. Mg^{2+}, with 12 protons, is the smallest.

QUICK CHECK

Why does graphite conduct electricity but diamond does not?

- Graphite contains ions that can move between the layers.
- In graphite one electron per carbon is delocalised and can move along the layers; in diamond all outer electrons are in bonds.
- The layers in graphite slide over each other.
- Diamond has stronger covalent bonds, which trap the electrons.

Conduction needs mobile charged particles. Each carbon in graphite uses only three electrons in bonds, and the fourth is delocalised within its layer. Sliding layers explain why graphite is soft, not why it conducts. Graphite contains no ions.

QUICK CHECK

The H–O–H bond angle in water is 104.5°, smaller than 109.5°. Why?

- Oxygen is more electronegative than hydrogen.
- Water forms hydrogen bonds.
- Water has only two bonding pairs, which repel each other less.
- The two lone pairs on oxygen repel more strongly than the bonding pairs and push the O–H bonds together.

Oxygen has four electron pairs, arranged roughly tetrahedrally. Lone pairs repel more than bonding pairs, and two lone pairs squeeze the angle by about 5° from 109.5°. Electronegativity affects the polarity of the bonds, not the angle.

QUICK CHECK

In the ammonium ion, NH_4^+, where do the two electrons of the dative bond come from?

- One from nitrogen and one from the hydrogen ion
- Both from the lone pair on nitrogen
- Both from the hydrogen ion
- From the other three N–H bonds

An H^+ ion has no electrons at all, so both electrons in the new bond come from nitrogen's lone pair. Once formed, the four N–H bonds are identical and the ion is tetrahedral.

QUICK CHECK

A grey solid melts at 1083 °C, conducts electricity as a solid and when molten, and is insoluble in water. What is its structure?

- giant ionic
- giant covalent
- giant metallic
- simple molecular

Conducting as a solid points to delocalised electrons, and conducting when molten as well, with a high melting temperature, fits a metal (this is copper). An ionic solid does not conduct until it melts.

AS · Unit 1 · WCH11 Structure, Bonding and Introduction to Organic Chemistry

Introductory organic chemistry and alkanes

Organic chemistry is the chemistry of carbon compounds, and there are millions of them. This topic gives you the tools to handle them all: how to draw and name organic molecules, how isomers arise, and how chemists describe the way bonds break. Then it puts those tools to work on the simplest family, the alkanes: where they come from, why they make good fuels, what pollution they cause, and how they react with halogens by a free-radical chain reaction.

Hazards and risks

Many organic compounds catch fire easily or are toxic, so before any practical you assess what could go wrong. Two words are used in a precise way, and the exam expects you to keep them apart.

- **Hazard:** a substance or a procedure that has the potential to cause harm. Bromine is a hazard because it is toxic and corrosive, whatever you do with it.
- **Risk:** the chance that harm will actually happen when the hazardous substance or procedure is used in a particular way. The risk depends on how much you use, how concentrated it is and what precautions you take.

So the hazard of a chemical is fixed, but the risk is something you can change. A dilute solution, or a small amount, carries a much lower risk than the pure substance, even though the hazard is the same. Risk can be reduced by:

- working on a smaller scale, or with more dilute solutions;
- taking precautions matched to the particular hazard (see the table);
- choosing a different method or a less hazardous substance that does the same job.

Hazard	Example in the lab	Typical precaution
Flammable	hexane, ethanol	no naked flames nearby; heat with a water bath or electric heater
Toxic	bromine, methanol	use a fume cupboard; wear gloves; avoid skin contact
Corrosive	concentrated acids and alkalis	wear goggles and gloves
Irritant	dilute acids and alkalis	wear goggles
Harmful	many organic liquids	avoid breathing vapour or swallowing; wash hands afterwards
Oxidising	potassium manganate(VII)	keep away from flammable and easily oxidised materials
Carcinogenic	some halogenoalkanes	replace with a safer substance where possible; fume cupboard; gloves

EXAM TIP

In a risk-assessment question, name the specific hazard and give a precaution that deals with **that** hazard. "Wear goggles" is not a sensible precaution for a flammable liquid; "keep away from naked flames" is.

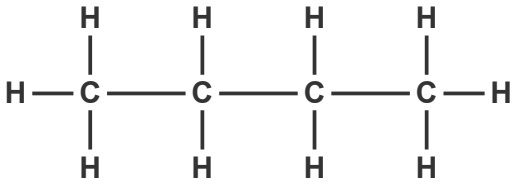
Formulae of organic compounds

Carbon forms four covalent bonds and can bond to other carbon atoms to make chains and rings of almost any length. That is why there are so many organic compounds. Start with two basic words:

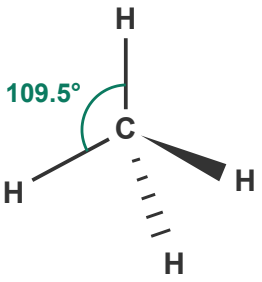
- **Hydrocarbon:** a compound made of hydrogen and carbon **only**.
- **Saturated:** contains only single carbon–carbon bonds. **Unsaturated:** contains at least one C=C double bond (or a triple bond).

The same molecule can be written in several ways, each showing a different amount of detail. You need to be able to read and write all of them.

Type of formula	What it shows	Butane
Molecular	the actual number of atoms of each element in one molecule	C_4H_{10}
Empirical	the simplest whole-number ratio of atoms of each element	C_2H_5
General	an algebraic formula that fits every member of a homologous series	C_nH_{2n+2} (all alkanes)
Structural	the minimum detail needed to show how the atoms are arranged, carbon by carbon, with no bonds drawn	$CH_3CH_2CH_2CH_3$ or $CH_3(CH_2)_2CH_3$
Displayed	every atom and every covalent bond	see the diagram below
Skeletal	only the carbon skeleton (as lines) and the functional groups; carbon atoms and the hydrogens on them are left out	a zig-zag of three lines



displayed formula of butane

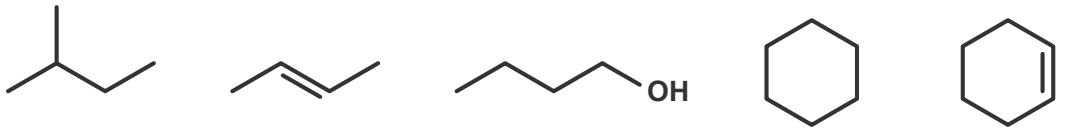


methane in 3D

Left: a displayed formula shows every bond, but drawn flat it makes the molecule look square. Right: around each saturated carbon the four bonds point to the corners of a tetrahedron, 109.5° apart. A wedge comes towards you, a hashed line goes away from you, and plain lines are in the plane of the page.

Skeletal formulae

Skeletal formulae are quick to draw and easy to read once you are used to them. Each line is a C–C bond, a carbon atom sits at every corner and at the end of every line, and you assume each carbon has enough hydrogens to make four bonds. Atoms other than C and H, and the hydrogens attached to them (as in –OH), are always written in.



2-methylbutane · but-2-ene · butan-1-ol · cyclohexane · cyclohexene

Five skeletal formulae. In but-2-ene the second line marks the C=C double bond. Cyclohexane, C_6H_{12}, is a ring of six CH_2 groups.

COMMON MISTAKE

When you draw a displayed formula, count the bonds round every carbon: there must be exactly four. Missing hydrogens and five-bonded carbons are the commonest ways to lose a mark. In a displayed formula, the O–H bond of an alcohol must be drawn too.

Homologous series and functional groups

- **Homologous series:** a family of compounds with the same functional group and the same general formula. Each member differs from the next by a CH_2 group.
- **Functional group:** the atom or group of atoms that gives a compound its characteristic chemical reactions. Molecules with the same functional group react in the same ways.

Because the functional group decides how a molecule reacts, all members of a series have **similar chemical properties**. Their **physical properties**, such as boiling temperature, change **gradually** along the series, because each extra CH_2 adds

eight electrons and makes the molecule a little longer, so the London forces between molecules get a little stronger each time.

Series	Functional group	Name ending or prefix	General formula	Example
Alkanes	C–C and C–H only	-ane	C_nH_{2n+2}	propane, $CH_3CH_2CH_3$
Alkenes	C=C	-ene	C_nH_{2n}	propene, $CH_2{=}CHCH_3$
Halogenoalkanes	–X (X = F, Cl, Br, I)	prefix fluoro-, chloro-, bromo-, iodo-	$C_nH_{2n+1}X$	1-bromopropane, $CH_3CH_2CH_2Br$
Alcohols	–OH	-ol (prefix hydroxy-)	$C_nH_{2n+1}OH$	propan-1-ol, $CH_3CH_2CH_2OH$
Aldehydes	–CHO (C=O at the end of a chain)	-al	$C_nH_{2n}O$	propanal, CH_3CH_2CHO
Ketones	C=O inside a chain	-one	$C_nH_{2n}O$	propanone, CH_3COCH_3
Carboxylic acids	–COOH	-oic acid	$C_nH_{2n+1}COOH$	propanoic acid, CH_3CH_2COOH

KEY IDEA

Aldehydes and ketones share the general formula $C_nH_{2n}O$, and alkenes share C_nH_{2n} with cycloalkanes. The same general formula does not always mean the same series: the functional group decides.

Naming organic compounds

Every name is built from three parts: a **stem** that says how many carbons are in the longest chain, an **ending** (suffix) for the main functional group, and **prefixes** for branches and other groups, each with a number to say where it is.

Carbons	1	2	3	4	5	6	7	8	9	10
Stem	meth	eth	prop	but	pent	hex	hept	oct	non	dec
Branch (alkyl group)	methyl, $–CH_3$	ethyl, $–C_2H_5$	propyl, $–C_3H_7$	a branch is named from the stem with -yl on the end						

The rules

1. Find the **longest continuous carbon chain**. It may bend round corners in the drawing. This gives the stem.
2. Number the chain from the end that gives the **main functional group** the lowest number. If there is no functional group needing a suffix, number from the end that gives the branches the lowest numbers.
3. Name each branch or substituent as a prefix with its position number. Use **di-, tri-, tetra-** when the same group appears more than once, and give every position (2,2-dimethyl, not 2-dimethyl).
4. Put different prefixes in **alphabetical order**, ignoring di-, tri- and so on (bromo before chloro before methyl).
5. Punctuation: **commas** between numbers, **hyphens** between numbers and words.
6. Leave out numbers that cannot vary: propanone, ethanoic acid and methylpropane need no numbers, because there is only one possible position.

Naming a branched alkane

1. Name $CH_3CH(CH_3)CH_2CH(C_2H_5)CH_3$. Draw it out first: a chain of five carbons, with a methyl on carbon 2 and an ethyl on carbon 4.
2. Look for the longest chain. Following the chain into the ethyl group gives **six** carbons, not five. So the stem is **hex** and the compound is a hexane.

   The ethyl group was never really a branch: it is part of the longest chain. A "branch" of two or more carbons on the second-to-last carbon is a warning sign that you have missed the longest chain.
3. Along this six-carbon chain there are now two methyl branches. Numbering from one end puts them on carbons 2 and 4; from the other end, on 3 and 5. Choose the lower set, 2 and 4.
4. Two methyl groups means "dimethyl", and both positions are given: **2,4-dimethylhexane**.

Names with functional groups

- **Halogenoalkanes:** the halogen is always a prefix: 2-chloropropane, 1-bromo-2-methylpropane. With two different halogens, list them alphabetically: $CH_2ClCHBrCH_3$ is **2-bromo-1-chloropropane** (numbering 1,2 beats 2,3).
- **Alkenes:** the position of the double bond is the lower-numbered of its two carbons: $CH_3CH{=}CHCH_3$ is but-2-ene. Two double bonds make a diene, and an "a" is added to the stem: penta-1,3-diene.
- **Alcohols:** the position number goes just before -ol: butan-2-ol, 2-methylpropan-2-ol. Two or three –OH groups make a diol or triol: ethane-1,2-diol, propane-1,2,3-triol.
- **Aldehydes and carboxylic acids:** the functional group is always at the end of the chain, so its carbon is carbon 1 and needs no number: butanal, 2-methylpropanoic acid. An acid group at both ends gives a dioic acid: ethanedioic acid, HOOC–COOH.
- **Ketones:** the C=O is inside the chain. From five carbons up you need a number: pentan-2-one and pentan-3-one are different compounds.

KEY IDEA

When to keep the "e" of the stem. If the suffix starts with a vowel, drop the final e of the alkane name: propan**ol**, ethan**oic** acid, butan**al**. If the suffix starts with a consonant (including di- and tri-), keep the e: ethan**e**-1,2-diol,

propan**e**dioic acid, pentan**e**-2,4-dione.

When a molecule has more than one functional group, the one highest in this order takes the suffix and sets the numbering. The others become prefixes (–OH becomes **hydroxy-**), except that a C=C keeps its "-en-" inside the name.

carboxylic acid > aldehyde > ketone > alcohol > alkene > halogenoalkane

Structural formula	Name	Why
$CH_3CH(OH)CH_2CH(CH_3)CH_3$	4-methylpentan-2-ol	number from the end nearer the –OH, so it gets 2, not 4
$CH_2=CHCH_2CH(OH)CH_3$	pent-4-en-2-ol	the alcohol outranks the alkene, so –OH gets the lower number
$CH_3CH(OH)COOH$	2-hydroxypropanoic acid	the acid takes the suffix and carbon 1; –OH becomes a prefix
$CH_3CH(OH)CH_2CHO$	3-hydroxybutanal	the aldehyde carbon is carbon 1
$CH_2=CHCOOH$	propenoic acid	the C=C can only be between carbons 2 and 3, so no number is needed

Structural isomers

Structural isomers have the same molecular formula but a different structural formula: the same atoms are joined up in a different order. Isomers are different compounds with different properties. Structural isomerism can arise in three ways, although the Edexcel specification simply calls them all structural isomers.

- **Chain isomers:** the carbon skeleton is different (straight or branched in different ways).
- **Position isomers:** the same functional group sits at a different position on the same skeleton.
- **Functional group isomers:** the atoms are arranged to give a different functional group altogether.

Chain isomers, C_5H_{12}



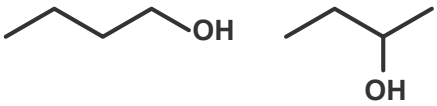
pentane, 36 °C

2-methylbutane, 28 °C



2,2-dimethylpropane, 10 °C

Position isomers, $C_4H_{10}O$

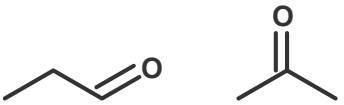


butan-1-ol

butan-2-ol

both $C_4H_{10}O$: –OH on a different carbon

Functional group isomers



propanal

propanone

both C_3H_6O: aldehyde and ketone

Chain isomers have different skeletons (and, as you will see below, different boiling temperatures). Butan-1-ol and butan-2-ol differ only in where the –OH is. Propanal and propanone have the same formula, C_3H_6O, but different functional groups.

Two more functional group pairs worth knowing: ethanol, CH_3CH_2OH, and methoxymethane, CH_3OCH_3, are both C_2H_6O; and hex-1-ene, $CH_2{=}CHCH_2CH_2CH_2CH_3$, and cyclohexane are both C_6H_{12}, because a ring uses up the same two hydrogens as a double bond.

EXAM TIP

To find all the isomers of a formula, work systematically: start with the longest chain, then shorten it by one carbon and try the branch in every position, and so on. Check every candidate by naming it. Two drawings with the same name are the same compound, not two isomers.

A second kind of isomerism, **stereoisomerism** (same structural formula, different arrangement in space), turns up with alkenes in the next topic.

Types of reaction and how bonds break

Classifying reactions

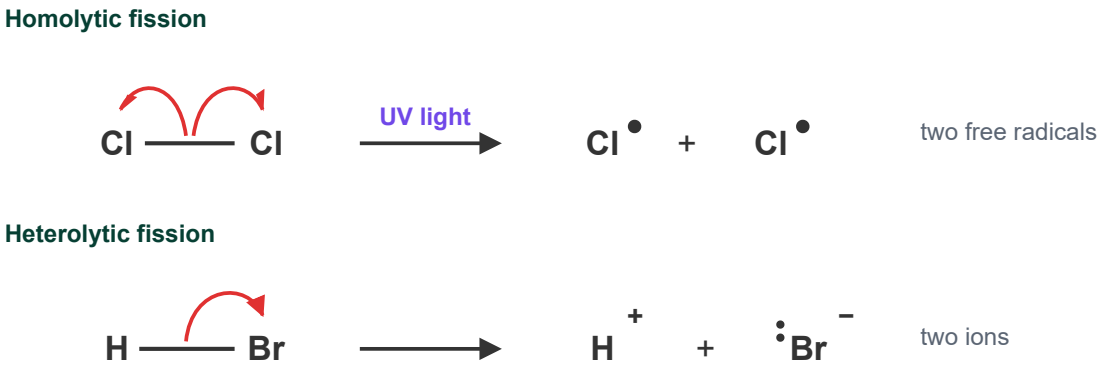
Type	What happens	Example
Addition	two molecules join to make a single product	ethene + bromine → 1,2-dibromoethane
Substitution	one atom or group is replaced by another	methane + chlorine → chloromethane + hydrogen chloride

Elimination	a small molecule (such as H_2O or HBr) is removed from a larger one, leaving a C=C	bromoethane → ethene + HBr
Oxidation	in organic chemistry, gain of oxygen or loss of hydrogen	ethanol → ethanal
Reduction	gain of hydrogen or loss of oxygen	ethene + hydrogen → ethane
Hydrolysis	a molecule is split by reaction with water	bromoethane + water → ethanol + HBr
Polymerisation	many small monomer molecules join into a long chain	ethene → poly(ethene)

Homolytic and heterolytic fission

A covalent bond is a shared pair of electrons. When it breaks, those two electrons can be shared out in two ways, and the two ways lead to completely different kinds of reaction.

- **Homolytic fission:** each atom takes **one** electron from the bond. This makes two **free radicals**. It is shown with half-headed ("fish-hook") arrows, each moving a single electron.
- **Heterolytic fission:** one atom takes **both** electrons. This makes a positive ion and a negative ion. It is shown with a full curly arrow, which moves a pair of electrons. The more electronegative atom usually takes the pair.



Homolysis of Cl–Cl gives two chlorine radicals, each with an unpaired electron (the dot). Heterolysis of H–Br gives H^+ and Br^-: bromine is more electronegative, so it keeps the bonding pair, which becomes a lone pair on the bromide ion.

Three kinds of attacking species

- **Free radical:** a species with an **unpaired electron**, shown by a dot, such as Cl• or $\bullet CH_3$. Radicals have no charge but are very reactive, because the unpaired electron "wants" a partner.
- **Electrophile:** an **electron-pair acceptor**. Electrophiles are attracted to regions of high electron density, such as a C=C bond. They are positive ions or have a δ+ atom: H^+, $Br^{\delta+}$ in a polarised Br_2 molecule.
- **Nucleophile:** an **electron-pair donor**. A nucleophile has a lone pair it can use to form a new covalent bond with a δ+ carbon: OH^-, CN^-, NH_3, H_2O.

KEY IDEA

Curly arrows show where electrons move. A full arrow moves a **pair** of electrons and always starts at a source of electrons: a **lone pair** or the middle of a **bond**. It ends where the pair goes: between two atoms to make a new bond, or on an atom that takes the pair when a bond breaks. A half-headed arrow moves one electron.

Alkanes: structure and physical properties

Alkanes are **saturated hydrocarbons** with the general formula C_nH_{2n+2}. Every carbon is tetrahedral, with bond angles of 109.5°, so a "straight" chain is really a zig-zag. Cycloalkanes, such as cyclohexane, are rings of CH_2 groups with the general formula C_nH_{2n}.

Carbon and hydrogen have very similar electronegativities, so C–H bonds are almost non-polar, and alkane molecules are **non-polar**. The only forces between alkane molecules are **London forces** (instantaneous dipole–induced dipole forces, see Topic 7). When an alkane boils, it is these weak forces that are overcome. The strong covalent bonds inside the molecules do not break. Two factors control how strong the London forces are:

- **Chain length.** Each extra CH_2 adds 8 electrons. More electrons give larger instantaneous dipoles, so the London forces are stronger and more energy is needed to separate the molecules: the boiling temperature rises steadily along the series.
- **Branching.** Isomers have the same number of electrons, but a branched molecule is more compact, closer to a sphere. Neighbouring molecules cannot lie alongside each other, so there is a smaller area of contact and fewer points where the molecules are close. The London forces are weaker and the boiling temperature is lower.

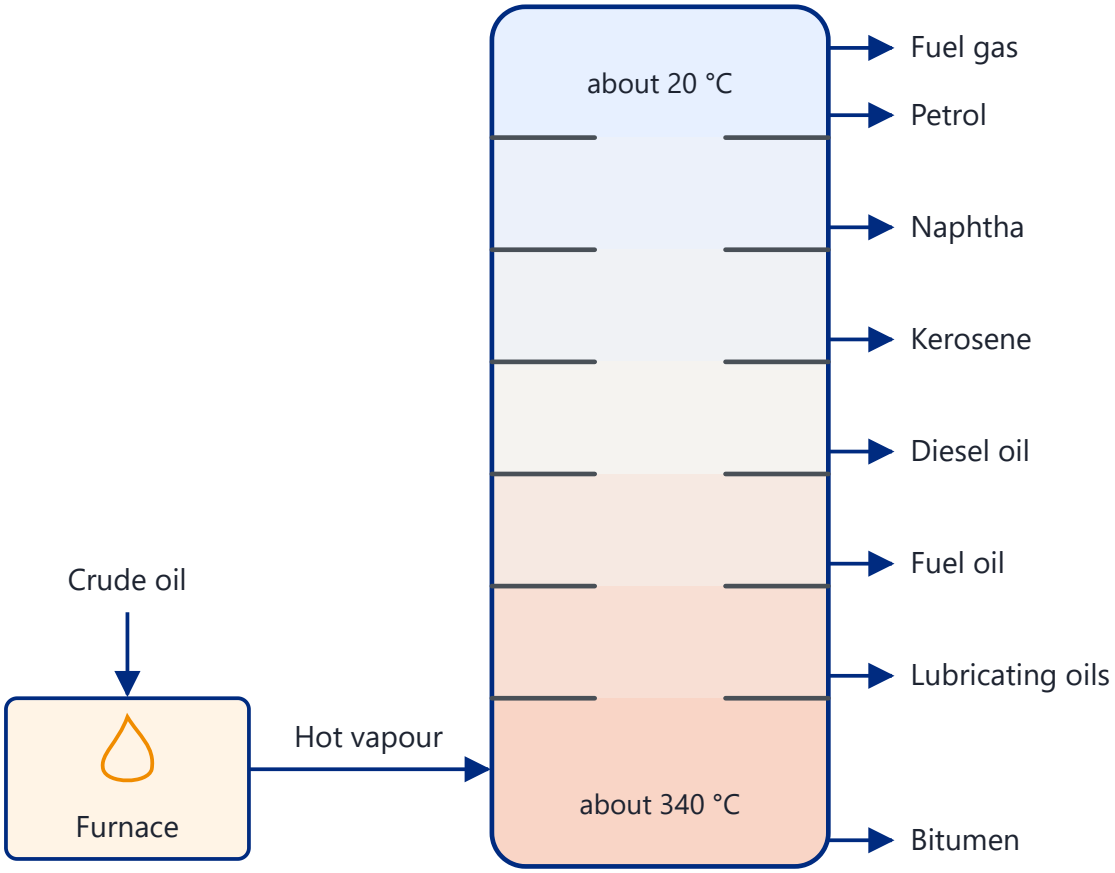
EXAM TIP

For an "explain the difference in boiling temperatures" answer about isomers, make three points: (1) same number of electrons, so any difference is not due to that; (2) the straight-chain molecules have a larger surface area of contact, so stronger London forces; (3) more energy is needed to overcome them, so the boiling temperature is higher. Always say "London forces **between** molecules": the covalent bonds do not break.

Alkanes are not very reactive. Their C–C and C–H bonds are strong (about 347 and 413 kJ mol^{-1}), and because the bonds are almost non-polar there are no δ+ or δ− sites to attract nucleophiles or electrophiles. Alkanes do two important things: they burn, and they react with halogens in ultraviolet light.

Crude oil and fractional distillation

Crude oil (petroleum) is a mixture of hydrocarbons, mostly alkanes, of many different chain lengths. On its own it is not much use. It is separated into **fractions**: mixtures of hydrocarbons with similar chain lengths and therefore similar boiling temperatures.



A fractionating column. The temperature falls from bottom to top; each fraction condenses and is drawn off at the level where the temperature has dropped below its boiling range.

How fractional distillation separates crude oil

1. The crude oil is heated in a furnace until most of it vaporises, and the hot vapour is fed in near the bottom of the column.
2. There is a temperature gradient up the column: it is hottest at the bottom and coolest at the top. The vapours rise through trays with bubble caps and cool as they go.
3. Each hydrocarbon condenses when it reaches a tray cooler than its boiling temperature, and the liquid collected there is drawn off as a fraction.
4. Small molecules have few electrons and weak London forces, so they have low boiling temperatures and rise to the top before they condense (or leave as gas). Large molecules have strong London forces and condense near the bottom. The heaviest part never vaporises and leaves at the bottom as a residue (bitumen).

This is a physical process: only the London forces between molecules are overcome and re-formed. No covalent bonds break.

Fraction	Approximate chain length	Main use
Refinery gas	C_1–C_4	bottled gas for heating and cooking
Petrol (gasoline)	C_5–C_{10}	fuel for cars
Naphtha	C_8–C_{12}	feedstock for the chemical industry
Kerosene (paraffin)	C_{10}–C_{16}	jet fuel
Diesel (gas oil)	C_{14}–C_{20}	fuel for diesel engines
Fuel oil and lubricating oil	C_{20}–C_{40}	ships and power stations; lubricants and waxes
Bitumen	above C_{40}	road surfaces and roofing

Cracking and reforming

Fractional distillation produces more of the long-chain fractions than anyone wants, and not enough of the short-chain fractions such as petrol, which are in high demand. Two processes fix the mismatch and also make more useful molecules.

Cracking

Cracking breaks C–C bonds in long-chain alkanes to make shorter alkanes and alkenes (sometimes hydrogen as well). Because strong covalent bonds have to be broken, cracking is a **chemical** change and needs a lot of energy. Industry mostly uses **catalytic cracking**, with a zeolite catalyst at around 450 °C; thermal cracking uses much higher temperatures and pressures instead.

Why crack? The shorter alkanes are used in petrol, and they are worth more than the long-chain fractions. The alkenes are even more valuable: unlike alkanes they are reactive, so they are the starting materials for polymers such as poly(ethene) and for chemicals such as ethanol and ethane-1,2-diol.

$$\mathrm{C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4} \qquad \mathrm{C_{16}H_{34} \rightarrow C_8H_{18} + C_6H_{12} + C_2H_4}$$

There is no single "right" answer when you crack an alkane: the bonds break at random. In the exam you will be told one product, or the products, and asked to balance the equation. Check it by counting carbons and hydrogens on each side. The alkane products have 2n + 2 hydrogens, the alkenes 2n.

Reforming

Reforming turns straight-chain alkanes into **branched alkanes** and **cyclic hydrocarbons** (cycloalkanes and arenes, such as benzene), usually over a platinum catalyst. Branched and cyclic hydrocarbons burn more smoothly in a car engine and are less likely to ignite too early (knocking), so they give a more efficient fuel with a higher octane number. Reforming to a ring releases hydrogen:

$$\mathrm{C_6H_{14} \rightarrow C_6H_{12} + H_2} \qquad \mathrm{C_6H_{14} \rightarrow C_6H_6 + 4H_2}$$

The first equation makes cyclohexane, the second benzene. Turning heptane into 2,2,3-trimethylbutane is also reforming: both are C_7H_{16}, so nothing else is produced.

Alkanes as fuels: combustion

A **fuel** is a substance that releases heat energy when it burns. Alkanes make good fuels: they are easy to transport, and their combustion is very exothermic, because the bonds made in CO_2 and H_2O are much stronger than the bonds broken in the alkane and in oxygen.

Complete combustion

With plenty of oxygen, the carbon ends up as carbon dioxide and the hydrogen as water.

Balancing the complete combustion of octane

1. Write the formulae: $C_8H_{18} + O_2 \rightarrow CO_2 + H_2O$.
2. Balance carbon first: 8 C atoms make **$8CO_2$**.
3. Balance hydrogen next: 18 H atoms make **$9H_2O$**.
4. Balance oxygen last, because O_2 is an element and you can use any amount of it. On the right: 8 × 2 + 9 = 25 O atoms, which is 12½ O_2.

$$\mathrm{C_8H_{18}(l) + 12\tfrac{1}{2}O_2(g) \rightarrow 8CO_2(g) + 9H_2O(l)}$$

Doubling everything gives whole numbers: $2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$. In general, C_nH_{2n+2} needs $(3n + 1)/2$ moles of O_2.

Incomplete combustion

If oxygen is limited, as in a poorly ventilated gas fire or inside a car engine, some carbon ends up as **carbon monoxide** or as **carbon** (soot) instead. Less energy is released per mole of fuel, because the carbon is not fully oxidised.

$$\mathrm{CH_4(g) + 1\tfrac{1}{2}O_2(g) \rightarrow CO(g) + 2H_2O(l)} \qquad \mathrm{CH_4(g) + O_2(g) \rightarrow C(s) + 2H_2O(l)}$$

KEY IDEA

Why carbon monoxide is so dangerous

CO is colourless and odourless, so you cannot tell it is there. It binds to the haemoglobin in red blood cells more strongly than oxygen does, so the haemoglobin can no longer carry oxygen around the body. A build-up from a faulty heater in a closed room can kill.

Pollution from burning fuels

Pollutant	Where it comes from	Effect
Carbon monoxide, CO	incomplete combustion	toxic: stops haemoglobin carrying oxygen
Carbon (soot) and other particulates	incomplete combustion, especially in diesel engines	breathing problems and lung disease; global dimming (particles reflect sunlight)
Unburnt hydrocarbons	fuel that passes through an engine without burning	help to form photochemical smog; some are carcinogenic
Oxides of nitrogen, NO and NO_2 (NO_x)	nitrogen and oxygen from the air react at the high temperature and spark inside an engine	toxic; NO_2 forms acid rain and smog
Sulfur dioxide, SO_2	sulfur impurities in fuels, especially coal and heavy oil	acid rain; breathing problems
Carbon dioxide, CO_2	all combustion of carbon-based fuels	greenhouse gas: contributes to global warming and climate change

Nitrogen oxides. The N≡N triple bond is very strong, so N_2 normally does not react with oxygen. In an engine the temperature is so high, and the spark supplies so much energy, that some does:

$$\mathrm{N_2(g) + O_2(g) \rightarrow 2NO(g)} \qquad \mathrm{2NO(g) + O_2(g) \rightarrow 2NO_2(g)}$$

Sulfur dioxide. Sulfur in the fuel burns to SO_2, which dissolves in rainwater and is oxidised to sulfuric acid, giving acid rain that damages buildings, forests and lakes. Most sulfur is now removed from petrol and diesel at the refinery.

$$\mathrm{S(s) + O_2(g) \rightarrow SO_2(g)}$$

Catalytic converters

A catalytic converter in a car exhaust removes CO, NO_x and unburnt hydrocarbons by turning them into CO_2, N_2 and H_2O. It is a ceramic honeycomb coated with a thin layer of **platinum, palladium and rhodium**. The honeycomb gives a very large surface area, so a small mass of these expensive metals catalyses a lot of gas.

$$\mathrm{2CO(g) + 2NO(g) \rightarrow 2CO_2(g) + N_2(g)}$$

$$\mathrm{C_8H_{18}(g) + 25NO(g) \rightarrow 8CO_2(g) + 12\tfrac{1}{2}N_2(g) + 9H_2O(g)}$$

COMMON MISTAKE

A catalytic converter does not remove carbon dioxide: it actually turns CO and hydrocarbons into more CO_2. Nor does it deal with sulfur dioxide.

Alternative fuels and carbon neutrality

Crude oil is a finite resource that took millions of years to form, so it will run out. **Biofuels** are fuels made from plants, which can be grown again each year, so they are **renewable**.

- **Bioethanol** is made by fermenting sugars from crops such as sugar cane or maize with yeast: $C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$.
- **Biodiesel** is made by reacting vegetable oils (for example from rapeseed) with methanol, using an alkali as a catalyst.

An activity is **carbon neutral** if it gives no overall increase in the amount of carbon dioxide in the atmosphere. Biofuels are often described as close to carbon neutral. Follow the carbon in bioethanol to see why.

Is bioethanol carbon neutral?

1. The crop takes in carbon dioxide as it grows, by photosynthesis:

$$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$$

   That is **6 CO_2 absorbed** for each glucose molecule.

2. Fermentation turns each glucose into two ethanol molecules and releases **2 CO_2**:

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$$

3. Burning the two ethanol molecules releases **4 CO_2**:

$$2C_2H_5OH + 6O_2 \rightarrow 4CO_2 + 6H_2O$$

4. Total released: 2 + 4 = 6 CO_2, exactly the amount absorbed. On paper the fuel is carbon neutral.

5. In practice it is not quite: energy, mostly from fossil fuels, is used to make fertiliser, run farm machinery, transport the crop and distil the ethanol. All of that releases extra CO_2.

   So biofuels reduce net CO_2 emissions compared with fossil fuels, but they are not completely carbon neutral.

Advantages of biofuels	Disadvantages of biofuels
renewable, so they will not run out	land used for fuel crops cannot grow food, which can push food prices up
much lower net CO_2 emissions than fossil fuels	forests may be cleared to make farmland
save crude oil for use as a chemical feedstock	not fully carbon neutral (fertilisers, machinery, transport)
no risk of large oil spills	older engines may need modifying to run on them

Free-radical substitution of alkanes

In the dark, methane and chlorine do not react. In **ultraviolet light** (or sunlight) they react at room temperature, and a hydrogen atom on the methane is replaced by a chlorine atom:

$$\mathrm{CH_4(g) + Cl_2(g) \xrightarrow{UV\ light} CH_3Cl(g) + HCl(g)}$$

This is a **substitution**, and because it happens through free radicals it is called **free-radical substitution**. Bromine reacts with alkanes in exactly the same way. The mechanism has three stages.



each chlorine atom keeps one electron from the shared pair

The mechanism: methane and chlorine

1. **Initiation.** UV light provides the energy to break the Cl–Cl bond by homolytic fission, making two chlorine radicals:

$$\mathrm{Cl_2 \rightarrow 2Cl^{\bullet}}$$

Cl–Cl is the weakest bond present (242 kJ mol^{-1}, against 413 kJ mol^{-1} for C–H), so UV breaks it and leaves the C–H bonds alone.

2. **Propagation, step 1.** A chlorine radical takes a hydrogen atom from a methane molecule, forming HCl and leaving a methyl radical:

$$\mathrm{Cl^{\bullet} + CH_4 \rightarrow HCl + {^{\bullet}CH_3}}$$

3. **Propagation, step 2.** The methyl radical takes a chlorine atom from a Cl_2 molecule, forming the product and a new chlorine radical:

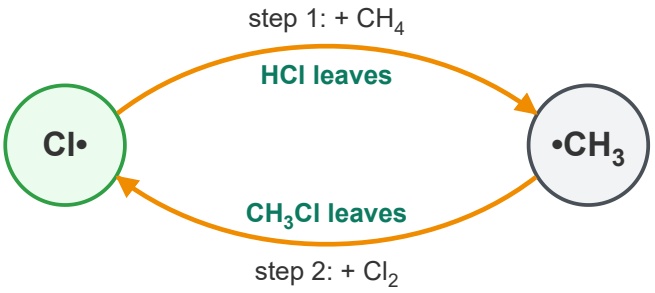
$$\mathrm{{^{\bullet}CH_3} + Cl_2 \rightarrow CH_3Cl + Cl^{\bullet}}$$

*Every propagation step has one radical on each side. The Cl• made in step 2 goes straight back into step 1, so one initiation can drive thousands of cycles: a **chain reaction**. Add the two propagation steps together and the radicals cancel, leaving the overall equation.*

4 **Termination.** Whenever two radicals meet, their unpaired electrons pair up to form a covalent bond, and no new radical is made. This removes radicals and stops chains:

$$Cl^{\bullet} + Cl^{\bullet} \rightarrow Cl_2 \qquad {}^{\bullet}CH_3 + Cl^{\bullet} \rightarrow CH_3Cl \qquad {}^{\bullet}CH_3 + {}^{\bullet}CH_3 \rightarrow CH_3CH_3$$

The last step makes ethane, an impurity in the product. Write it with structural formulae (CH_3CH_3) so it is clear the two carbons have joined.



the radical is regenerated every time round, so the cycle keeps going

The two propagation steps form a loop. Each time round, one CH_4 and one Cl_2 are used up, one HCl and one CH_3Cl are made, and the chlorine radical comes back out.

Applying the mechanism to other alkanes

The pattern is always the same: initiation splits the halogen; in propagation step 1 the halogen radical removes an H from the alkane, and in step 2 the alkyl radical takes a halogen atom from X_2. For bromine and ethane:

$$Br_2 \rightarrow 2Br^{\bullet}$$

$$Br^{\bullet} + CH_3CH_3 \rightarrow HBr + {}^{\bullet}CH_2CH_3 \qquad {}^{\bullet}CH_2CH_3 + Br_2 \rightarrow CH_3CH_2Br + Br^{\bullet}$$

$${}^{\bullet}CH_2CH_3 + Br^{\bullet} \rightarrow CH_3CH_2Br \qquad {}^{\bullet}CH_2CH_3 + {}^{\bullet}CH_2CH_3 \rightarrow CH_3CH_2CH_2CH_3$$

With propane, a hydrogen can be removed from an end carbon or from the middle carbon, so two different products form: 1-bromopropane and 2-bromopropane. To show substitution on the middle carbon, put the radical dot on that carbon:

$$CH_3CH_2CH_3 + Br^{\bullet} \rightarrow HBr + CH_3\overset{\bullet}{C}HCH_3 \qquad CH_3\overset{\bullet}{C}HCH_3 + Br_2 \rightarrow CH_3CHBrCH_3 + Br^{\bullet}$$

COMMON MISTAKE

Two common slips: making H_2 instead of HCl in propagation step 1 (the halogen radical always takes the hydrogen, so the side product is always the hydrogen halide), and putting the radical dot on the wrong carbon when you are asked for a particular isomer.

Further substitution and the limits of the reaction

The chloromethane made in the first cycle still has C–H bonds, so a chlorine radical can attack it too:

$$Cl^{\bullet} + CH_3Cl \rightarrow HCl + {}^{\bullet}CH_2Cl \qquad {}^{\bullet}CH_2Cl + Cl_2 \rightarrow CH_2Cl_2 + Cl^{\bullet}$$

Repeating this can give $CHCl_3$ and finally CCl_4. Overall equations add up all the substitutions, with HCl always as the other product, never H_2:

$$CH_4 + 2Cl_2 \rightarrow CH_2Cl_2 + 2HCl \qquad CH_4 + 4Cl_2 \rightarrow CCl_4 + 4HCl$$

KEY IDEA

Why free-radical substitution is a poor way to make one particular compound

It gives a **mixture of products**: further substitution (CH_2Cl_2, $CHCl_3$, CCl_4), substitution at different positions in longer chains (isomers), and termination products such as ethane or butane. The products have to be separated, and the yield of any one of them is low. Using a large **excess of the alkane** makes the monosubstituted product more likely, because a radical is then far more likely to meet an alkane molecule than a molecule that has already been substituted. An excess of halogen pushes the reaction towards full substitution.

Quick checks

QUICK CHECK

A teacher uses 0.1 mol dm^{-3} sodium hydroxide instead of 2 mol dm^{-3} sodium hydroxide in a class practical. What has changed?

- The hazard is lower but the risk is the same.
- Both the hazard and the risk are unchanged.
- The risk is lower; the hazard of sodium hydroxide itself is unchanged.
- The solution is no longer a hazard.

A hazard is a property of the substance, so sodium hydroxide is still corrosive or irritant. Using it more dilute reduces the chance and severity of harm, which is the risk.

QUICK CHECK

What is the empirical formula of hexane?

- C_6H_{14}
- C_3H_7 ✓
- CH_2
- C_nH_{2n+2}

Hexane is C_6H_{14}; dividing by the common factor 2 gives C_3H_7. C_6H_{14} is the molecular formula and C_nH_{2n+2} is the general formula.

QUICK CHECK

What is the correct name of $(CH_3)_2CHCH(CH_3)CH_2CH_3$?

- 2-methyl-3-methylpentane
- 3,4-dimethylpentane
- 2-ethyl-3-methylbutane
- 2,3-dimethylpentane ✓

The longest chain has five carbons, with methyl groups on carbons 2 and 3 when numbered from the end nearer the branches. Identical groups are combined with di- and every position is given. "2-ethyl…butane" misses the longest chain.

QUICK CHECK

Which pair of compounds are functional group isomers?

- propanal and propanone
- propan-1-ol and propan-2-ol
- butane and 2-methylpropane
- ethane and ethene

Both are C_3H_6O, but one is an aldehyde and the other a ketone. The propanols are position isomers, butane and methylpropane are chain isomers, and ethane and ethene have different molecular formulae, so they are not isomers at all.

QUICK CHECK

Pentane boils at 36 °C and 2,2-dimethylpropane at 10 °C. Which statement explains the difference?

- Pentane has more electrons per molecule.
- Pentane molecules have a larger surface area of contact, so the London forces between them are stronger.
- More covalent bonds must be broken to boil pentane.
- Pentane molecules are polar but 2,2-dimethylpropane molecules are not.

They are isomers, C_5H_{12}, so they have the same number of electrons. The difference comes from shape: the compact, branched molecule touches its neighbours over a smaller area. No covalent bonds break on boiling, and both are non-polar.

QUICK CHECK

Which equation correctly shows the complete combustion of propane?

- $C_3H_8 + 3½O_2 \rightarrow 3CO + 4H_2O$
- $C_3H_8 + 4O_2 \rightarrow 3CO_2 + 4H_2$
- $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 8H_2O$
- **$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$**

3 C give $3CO_2$; 8 H give $4H_2O$; the right-hand side then has 6 + 4 = 10 O atoms, which is $5O_2$. The first option is incomplete combustion (and does not balance), and $8H_2O$ would need 16 hydrogen atoms.

QUICK CHECK

Which of these is a propagation step in the reaction of ethane with chlorine in UV light?

- $Cl_2 \rightarrow 2Cl•$
- **$Cl• + C_2H_6 \rightarrow HCl + •C_2H_5$**
- $•C_2H_5 + Cl• \rightarrow C_2H_5Cl$
- $Cl• + C_2H_6 \rightarrow C_2H_5Cl + H•$

A propagation step has one radical on each side. The first option is initiation and the third is termination (two radicals in, none out). In the last option the chlorine radical would have to replace H directly, but the chlorine radical removes a hydrogen atom to form HCl.

QUICK CHECK

Why does a catalytic converter use a ceramic honeycomb coated with a thin layer of platinum and rhodium?

- To absorb carbon dioxide from the exhaust
- To raise the temperature of the exhaust gases
- To give a large surface area using only a small mass of expensive metal
- To filter out soot particles

The reactions happen on the surface of the metals, so a large area means fast conversion of CO, NO_x and hydrocarbons, and a thin coating keeps the cost down. The converter makes CO_2; it does not remove it.

Alkenes

Alkenes contain a carbon–carbon double bond, and that one bond changes everything. It makes alkenes far more reactive than alkanes, it stops the molecule twisting (which creates a new kind of isomer), and it lets small molecules join into the long chains of addition polymers. This topic explains all three, starting from what a double bond actually is.

Alkenes and their names

Alkenes are **unsaturated hydrocarbons**: they contain at least one C=C double bond. With one double bond and no rings, their general formula is C_nH_{2n}. Cycloalkenes, such as cyclohexene, C_6H_{10}, have a ring as well as a double bond, so they have two fewer hydrogens: C_nH_{2n-2}.

The two carbons of the C=C and the four atoms joined to them all lie in one plane, with bond angles of about 120°. Three regions of electron density around each carbon (two single bonds and the double bond) repel each other as far apart as possible.

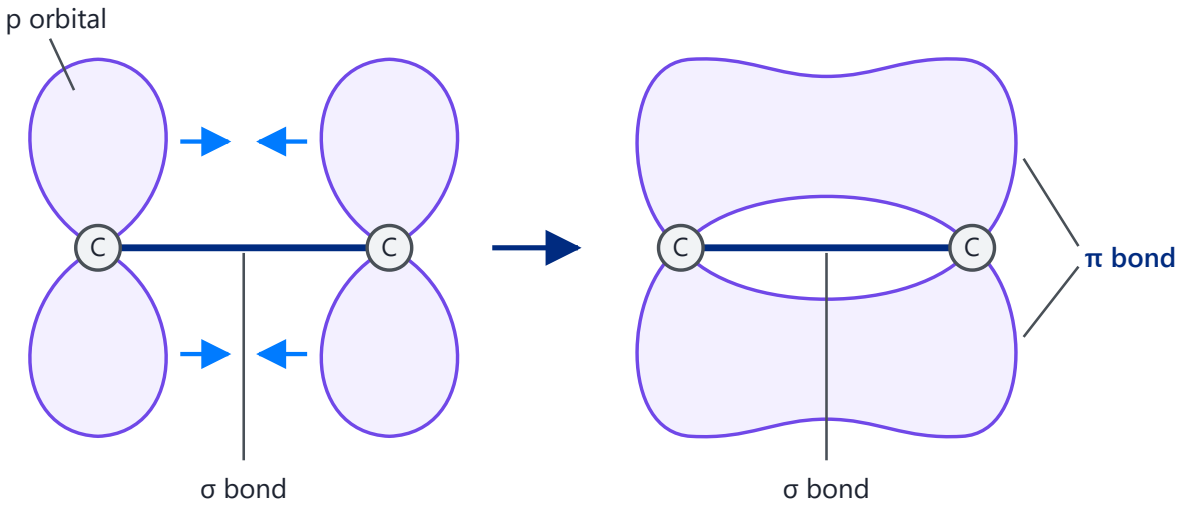
Name	Structural formula	Note
ethene	$CH_2=CH_2$	the simplest alkene
propene	$CH_2=CHCH_3$	no number needed: the C=C can only start at carbon 1
but-1-ene	$CH_2=CHCH_2CH_3$	position isomers: the number is the lower-numbered carbon of the C=C
but-2-ene	$CH_3CH=CHCH_3$	
2-methylpropene	$(CH_3)_2C=CH_2$	a chain isomer of the butenes
buta-1,3-diene	$CH_2=CHCH=CH_2$	two double bonds: a diene, with an extra "a" in the stem

The double bond: σ and π bonds

A covalent bond forms where two atomic orbitals overlap and share a pair of electrons. Orbitals can overlap in two different ways, and a double bond uses both.

- **σ bond (sigma):** the orbitals overlap end-on, directly along the line between the two nuclei. The shared electrons sit between the nuclei, where they attract both strongly. Every single bond is a σ bond.
- **π bond (pi):** two p orbitals, one on each carbon, overlap **sideways**. The shared electron density forms two regions, one above and one below the plane of the molecule, with none on the line between the nuclei.

A C=C double bond is **one σ bond plus one π bond**.



The p orbitals on neighbouring carbons overlap sideways, above and below the σ bond, to form the π bond.

Sideways overlap is less effective than end-on overlap, and the π electrons are further from the two nuclei, so **the π bond is weaker than the σ bond**. You can see this in the mean bond enthalpies: C=C is 612 kJ mol^{-1} but C–C is 347 kJ mol^{-1}, so the π bond adds only about 265 kJ mol^{-1}. Two big consequences follow.

- **Reactivity.** The π electrons are exposed above and below the molecule, making a region of high electron density that attracts **electrophiles** (electron-pair acceptors). The π bond is also relatively easy to break. So alkenes undergo **addition** reactions: the π bond breaks and two new σ bonds form.
- **Restricted rotation.** The two p orbitals only overlap when they are parallel. Twisting one end of the molecule would pull them out of line and break the π bond, so there is no free rotation about a C=C bond at room temperature. (Rotation about a C–C single bond is free, because end-on overlap does not depend on the angle.)

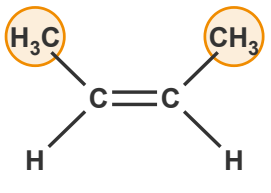
EXAM TIP

To count σ and π bonds, remember: every single bond is one σ bond, and every double bond is one σ plus one π. Propene, CH_2=$CHCH_3$, has six C–H bonds, one C–C and one C=C, so it has 8 σ bonds and 1 π bond.

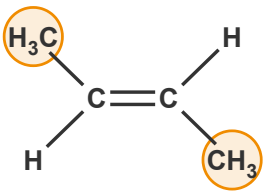
E/Z stereoisomerism

Stereoisomers have the same structural formula but a different arrangement of their atoms in space. Alkenes can show a kind called **E/Z isomerism** (also called geometric isomerism). It needs two things together:

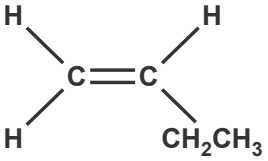
1. **restricted rotation** about the C=C double bond, so the arrangement is locked in; and
2. **two different groups on each carbon** of the double bond. If either carbon carries two identical groups, swapping them makes no difference, so there are no E/Z isomers.



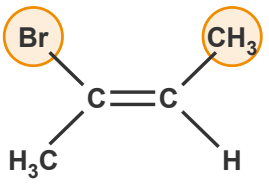
Z-but-2-ene
(cis-but-2-ene)



E-but-2-ene
(trans-but-2-ene)



but-1-ene
two H on one carbon: no E/Z



Z-2-bromobut-2-ene
Br and CH_3 on the same side

The higher-priority group on each carbon is circled. In but-1-ene the left-hand carbon carries two hydrogen atoms, so swapping them changes nothing and there are no E/Z isomers.

Naming E and Z isomers

To decide which isomer is which, give the two groups on each carbon a **priority** using the Cahn–Ingold–Prelog (CIP) rule: the group whose atom bonded directly to the C=C carbon has the **higher atomic number** has the higher priority. Then compare the two higher-priority groups:

- **Z isomer** (from the German *zusammen*, together): the higher-priority groups are on the **same side** of the double bond.
- **E isomer** (from *entgegen*, opposite): the higher-priority groups are on **opposite sides**.

Is the fourth molecule above E or Z?

1. Look at the left-hand carbon of the C=C. It carries Br and CH_3. The atoms bonded directly to the carbon are Br (atomic number 35) and C (atomic number 6). **Br has priority.**
2. Look at the right-hand carbon. It carries CH_3 and H: C (6) beats H (1), so **CH_3 has priority.**
3. Both priority groups, Br and the right-hand CH_3, are above the double bond: same side, so this is **Z-2-bromobut-2-ene**.

Notice that the two CH_3 groups are on opposite sides, so anyone using cis/trans by comparing the methyl groups would call it "trans". That clash is why the E/Z system, which works for any groups, is the one to use.

Cis–trans naming is a special case of E/Z that works when the two carbons carry a matching pair of groups, as in but-2-ene or 1,2-dichloroethene (CHCl=CHCl). Cis means the matching groups are on the same side; trans means opposite. For these molecules Z is the same as cis and E is the same as trans, because the matching groups (CH_3, or Cl) are also the higher-priority ones.

E/Z isomers are different compounds with different physical properties. Z-but-2-ene boils at about 4 °C and E-but-2-ene at about 1 °C. Their chemical reactions are very similar, because they have the same functional group.

COMMON MISTAKE

But-1-ene, CH_2=$CHCH_2CH_3$, has no E/Z isomers: carbon 1 carries two hydrogen atoms. It is, however, a structural isomer of but-2-ene. Always check both carbons of the C=C before you draw E/Z isomers.

Addition reactions of alkenes

In an **addition reaction** two molecules join to make one product. For an alkene, the π bond breaks and each carbon of the old C=C gains a new σ bond, so the product is saturated.

Reagent	Conditions	Product	Type of reaction
hydrogen, H_2	nickel catalyst, about 150 °C	alkane	addition (hydrogenation); reduction
halogen, e.g. Br_2	room temperature; bromine in an organic solvent or as bromine water	dihalogenoalkane	electrophilic addition
hydrogen halide, e.g. HBr	room temperature	halogenoalkane	electrophilic addition
steam, $H_2O(g)$	phosphoric acid catalyst, about 300 °C and 60–70 atm	alcohol	addition (hydration)
potassium manganate(VII), acidified	room temperature	diol	oxidation

$$\mathrm{CH_2{=}CH_2 + H_2 \xrightarrow{Ni,\ 150\ ^\circ C} CH_3CH_3} \qquad \mathrm{CH_2{=}CH_2 + Br_2 \rightarrow CH_2BrCH_2Br}$$

$$\mathrm{CH_2{=}CHCH_3 + HBr \rightarrow CH_3CHBrCH_3} \qquad \mathrm{CH_2{=}CH_2(g) + H_2O(g) \rightleftharpoons CH_3CH_2OH(g)}$$

$$\mathrm{CH_2{=}CH_2 + [O] + H_2O \rightarrow HOCH_2CH_2OH}$$

The products are 1,2-dibromoethane, 2-bromopropane (the main product, explained below), ethanol and ethane-1,2-diol. In the last equation [O] stands for oxygen supplied by the oxidising agent.

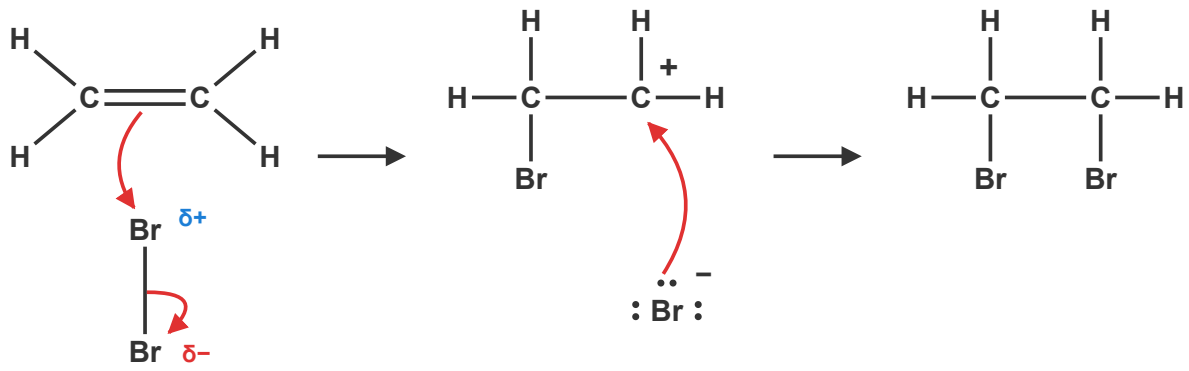
- **Hydrogenation** is used to turn liquid vegetable oils (which contain C=C bonds) into solid spreads such as margarine.
- **Hydration** of ethene is how most industrial ethanol is made. It is reversible, so the ethene that does not react is recycled. Every atom of the reactants ends up in the product, so the atom economy is 100%. The high pressure it needs makes it unsuitable for the school laboratory.

Electrophilic addition: the mechanism

An **electrophile** is an **electron-pair acceptor**. It is attracted to the high electron density of the π bond. The mechanism always has the same two steps: the electrophile accepts a pair of electrons from the π bond, which leaves a positively charged **carbocation** (a carbon with only three bonds and a positive charge); then a negative ion uses a lone pair to bond to that carbon.

Bromine and ethene

Bromine is non-polar, so how can it be an electrophile? As a Br_2 molecule approaches the alkene, the π electrons repel the electrons of the Br–Br bond towards the far bromine. This **induces a dipole**: the nearer bromine becomes δ+ and can accept a pair of electrons.



π electrons induce a dipole in Br_2 | carbocation + bromide ion | **1,2-dibromoethane**

Curly arrows (red) show electron pairs moving. The lone pairs drawn on Br^- are the ones that matter: the final arrow starts at one of them.

Mechanism: ethene + bromine → 1,2-dibromoethane

1. The π bond induces a dipole in Br_2: $Br^{\delta+}$–$Br^{\delta-}$.
2. A curly arrow from the π bond to $Br^{\delta+}$ shows the π electron pair forming a new C–Br bond. At the same time, a curly arrow from the Br–Br bond to $Br^{\delta-}$ shows that bond breaking **heterolytically**: the far bromine leaves as a bromide ion, Br^-.
3. That leaves a **carbocation intermediate**: the other carbon has lost its share of the π electrons and carries a positive charge.
4. A curly arrow from a **lone pair** on Br^- to C^+ shows the bromide ion forming the second C–Br bond. The product is 1,2-dibromoethane, CH_2BrCH_2Br.

Exam answers need: the δ+ and δ− on Br_2, the arrow from the π bond (the middle of the C=C), the arrow from the Br–Br bond, the structure of the carbocation with its + charge, a lone pair on Br^- with the final arrow starting from it.

Hydrogen bromide and alkenes

HBr is already polar, because bromine is more electronegative than hydrogen: $H^{\delta+}$–$Br^{\delta-}$. The mechanism is the same: the π bond attacks $H^{\delta+}$, the H–Br bond breaks heterolytically, and Br^- attacks the carbocation. With ethene the product is bromoethane. Hydrogen chloride reacts in just the same way.

Unsymmetrical alkenes: major and minor products

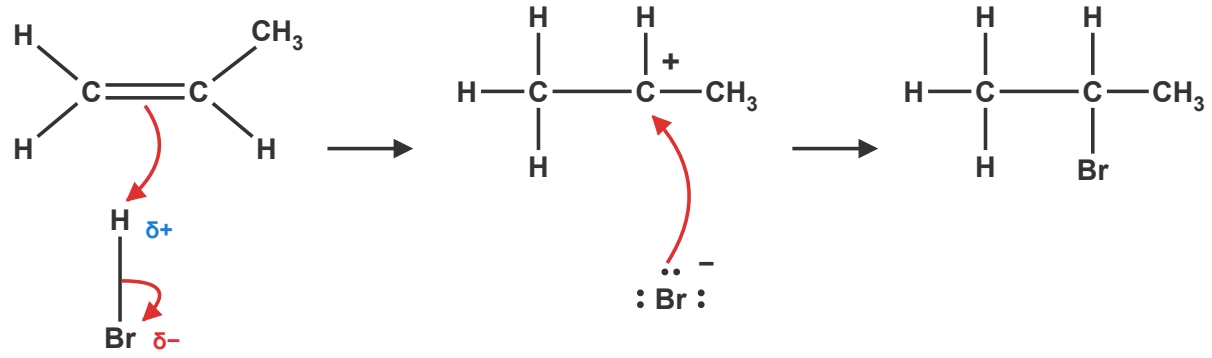
When HBr adds to propene, the hydrogen can go to either carbon of the C=C, so there are two possible carbocations and two possible products:

- H adds to the end carbon (CH_2): the positive charge is on the middle carbon, which is bonded to two other carbons. This is a **secondary** carbocation, and it leads to **2-bromopropane**.
- H adds to the middle carbon: the positive charge is on the end carbon, bonded to only one other carbon. This is a **primary** carbocation, and it leads to **1-bromopropane**.

KEY IDEA

Carbocation stability: tertiary > secondary > primary

Alkyl groups (CH_3, C_2H_5 and so on) are **electron-releasing**: they push electron density towards the positive carbon, which spreads out (reduces) the positive charge and stabilises the ion. The more alkyl groups on C^+, the more stable the carbocation. The more stable carbocation forms more easily and faster, so **the major product is the one formed via the more stable carbocation**.



H–Br is polar: Hδ+ is attacked

secondary carbocation (more stable)

2-bromopropane (major)

Propene and HBr by the major route. The H adds to the CH_2 end, so the positive charge sits on the middle carbon, which has two alkyl groups around it. The minor route puts the H on the middle carbon and goes through a primary carbocation to 1-bromopropane.

So propene and HBr give mainly 2-bromopropane, with a little 1-bromopropane. This pattern is sometimes summed up as "the hydrogen goes to the carbon that already has more hydrogens", but in the exam you must explain it in terms of carbocation stability.

EXAM TIP

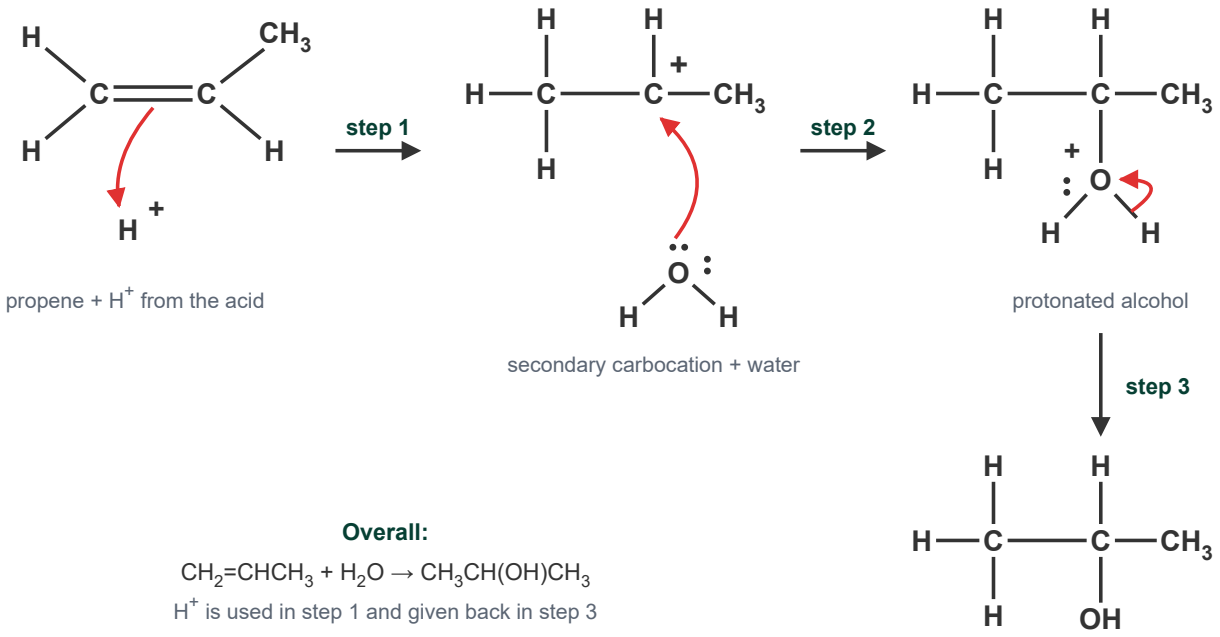
For an "explain why there is a major and a minor product" question: (1) draw or name both possible carbocations; (2) classify them as primary, secondary or tertiary; (3) state which is more stable, because it has more electron-releasing alkyl groups that reduce the positive charge; (4) say the major product forms via the more stable carbocation. If both carbocations are of the same type, expect roughly equal amounts of the two products.

EXAM TIP

Mixed halogens. A molecule such as BrCl is polar because chlorine is more electronegative: $Br^{\delta+}$–$Cl^{\delta-}$. So Br is the electrophile and adds first, and Cl^- attacks the carbocation. With propene, Br goes to the end carbon (giving the secondary carbocation) and the main product is 1-bromo-2-chloropropane.

Applying the mechanism: steam and an acid catalyst

Hydration follows the same pattern with H^+ (from the phosphoric acid) as the electrophile and a water molecule as the species that attacks the carbocation, using a lone pair on oxygen. The H^+ is given back at the end, which is why the acid is a catalyst. With propene, the secondary carbocation is favoured, so the main product is propan-2-ol.



Overall:

$CH_2{=}CHCH_3 + H_2O \rightarrow CH_3CH(OH)CH_3$

H^+ is used in step 1 and given back in step 3

+ H^+: the catalyst is regenerated

Three steps: H^+ is the electrophile; water attacks the carbocation with a lone pair on oxygen; then the extra H^+ is lost and the catalyst is regenerated.

Acid-catalysed hydration of propene

1. The π bond attacks H^+. H adds to the end carbon, giving the more stable secondary carbocation, $CH_3C^+HCH_3$.
2. A lone pair on the oxygen of a water molecule forms a bond to C^+. The oxygen now has three bonds and carries the positive charge.
3. An O–H bond breaks, with both electrons going to the oxygen, and H^+ is released. The product is propan-2-ol and the catalyst is regenerated.

$$CH_2{=}CHCH_3 + H_2O \xrightarrow{H^+} CH_3CH(OH)CH_3$$

Testing for a C=C double bond

Test	Result with an alkene	Result with an alkane
Shake with **bromine water**	orange (or yellow-brown) → **colourless**	stays orange
Shake with **acidified potassium manganate(VII)**	purple → **colourless**	stays purple

Both colours disappear because the coloured reagent is used up by addition across the C=C (bromine) or by oxidation of the C=C to a diol (manganate(VII)). Alkanes are saturated, so neither reaction can happen; at room temperature in the dark, bromine water does not react with them at all. In bromine water the water molecules, present in huge excess, compete with Br^- to attack the carbocation, so the main product with ethene is actually 2-bromoethanol:

$$CH_2{=}CH_2 + Br_2 + H_2O \rightarrow CH_2BrCH_2OH + HBr$$

COMMON MISTAKE

Say "decolourised" or "turns colourless". "Goes clear" is wrong: a coloured solution can be clear (transparent). And remember the test needs no UV light: in UV light, bromine would slowly react with an alkane too, by free-radical substitution.

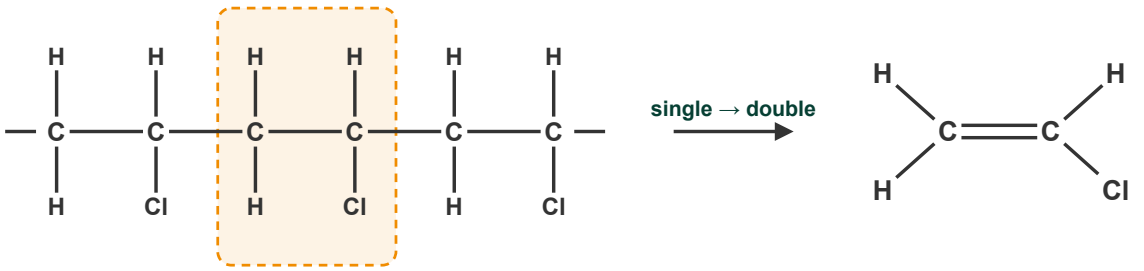
Addition polymers

Under the right conditions (high pressure, or a catalyst) alkene molecules join together into very long chains. The π bond of each **monomer** opens up and forms σ bonds to its neighbours, so thousands of monomers become one **polymer** molecule. Nothing else is produced, so this is **addition polymerisation**.

- **Monomer:** the small molecule (here an alkene) that joins up to form the polymer.
- **Repeat unit:** the smallest section of the chain that, repeated over and over, gives the whole polymer. For an addition polymer it is two carbon atoms long: the two carbons of the old C=C, now joined by a single bond.

$$n\,CH_2{=}CHCH_3 \rightarrow -[CH_2{-}CH(CH_3)]_n-$$

To draw the repeat unit, draw the monomer with its four groups around the C=C, change the double bond to a single bond, keep the four groups exactly where they were, and add bonds sticking out of each end through square brackets. The *n* goes outside the bracket, and in an equation in front of the monomer as well. The polymer is named by putting the monomer's name in brackets after "poly".



one repeat unit: two backbone carbons

chloroethene, the monomer

Working backwards: box any two neighbouring backbone carbons, keep their four groups, and put the C=C back.

To go backwards, from a polymer to its monomer, pick out a two-carbon repeat unit from the backbone, remove the end bonds and put the double bond back between the two carbons. The M_r of a repeat unit is the same as that of the monomer, so the number of repeat units in a chain is M_r(polymer) ÷ M_r(monomer). A poly(propene) chain of M_r 210 000 contains 210 000 ÷ 42.0 = 5000 repeat units.

Poly(alkenes) are saturated, with only strong, non-polar C–C and C–H bonds, so they are **unreactive**, just like alkanes. That makes them useful (they do not corrode or rot, and they are safe to store food in) but also creates a problem: they are **not biodegradable** and can last for hundreds of years.

Dealing with polymer waste

Method	Benefits	Drawbacks
Recycling: sort, clean, melt and remould	saves crude oil (the raw material) and the energy used to make new polymer; less waste in landfill	collecting and sorting is expensive in energy and labour; different polymers must be separated, because a mixture gives poor-quality material
Incineration with energy recovery	the heat released generates electricity; the volume of waste is greatly reduced	releases CO_2, a greenhouse gas; burning chlorine-containing polymers such as PVC produces toxic, acidic HCl
Feedstock for cracking	polymers are broken down into small molecules that can be made into new polymers and other chemicals	needs energy and suitable plant; waste must still be collected and sorted
Biodegradable polymers	made from renewable plant materials such as starch from maize; broken down by microorganisms	land is needed to grow the crops; many only break down quickly in the right conditions, such as industrial composting

Chemists reduce the harm from incineration by removing toxic waste gases before they reach the atmosphere. Acidic gases such as HCl are "scrubbed" by passing the waste gases through a base, such as calcium oxide or calcium carbonate:

$$\mathrm{CaO(s) + 2HCl(g) \rightarrow CaCl_2(s) + H_2O(l)} \qquad \mathrm{CaCO_3(s) + 2HCl(g) \rightarrow CaCl_2(s) + H_2O(l) + CO_2(g)}$$

Quick checks

QUICK CHECK

Which statement about the C=C bond in ethene is correct?

- It is two σ bonds, so it is twice as strong as a C–C bond.
- It is a σ bond and a π bond, and the π bond is the weaker of the two.
- The π bond lies along the line between the two carbon nuclei.
- The π bond allows free rotation about the C=C bond.

A double bond is one σ bond plus one π bond. The π bond comes from sideways overlap of p orbitals above and below the plane, which is less effective than end-on overlap, so it is weaker (612 is less than 2 × 347 kJ mol^{-1}). It prevents rotation rather than allowing it.

QUICK CHECK

Which compound has E/Z isomers?

- propene, $CH_2=CHCH_3$
- 2-methylpropene, $(CH_3)_2C=CH_2$
- 1,1-dichloroethene, $CCl_2=CH_2$
- pent-2-ene, $CH_3CH=CHCH_2CH_3$

In pent-2-ene each carbon of the C=C carries two different groups (H and CH_3; H and C_2H_5). In each of the others, one carbon of the C=C carries two identical groups.

QUICK CHECK

In CHBr=CHCl, the Br and the Cl are on opposite sides of the double bond. What is the isomer?

- E-1-bromo-2-chloroethene
- Z-1-bromo-2-chloroethene
- It has no E/Z isomers.
- It cannot be named, because Br and Cl are different groups.

On each carbon the halogen beats H (35 or 17 against 1), so Br and Cl are the priority groups. Opposite sides means E. Cis–trans would struggle here, but E/Z works for any groups.

QUICK CHECK

What is the major product when hydrogen bromide reacts with 2-methylpropene, $(CH_3)_2C=CH_2$?

- 1-bromo-2-methylpropane
- 2-bromo-2-methylpropane
- 1,2-dibromo-2-methylpropane
- 2-methylpropan-2-ol

H adding to the CH_2 carbon leaves a tertiary carbocation, $(CH_3)_3C^+$, stabilised by three electron-releasing methyl groups. Br^- adds to that carbon. The other route goes through a much less stable primary carbocation and gives the minor product, 1-bromo-2-methylpropane.

QUICK CHECK

In the mechanism of the reaction between ethene and bromine, where does the first curly arrow start?

- At a lone pair on the $Br^{\delta-}$ atom
- At the Br–Br bond
- At the C=C double bond (the π bond)
- At a C–H bond

The π electrons are the electron source: they move to form a bond to $Br^{\delta+}$. The arrow from the Br–Br bond is the second arrow, showing heterolytic fission. A lone pair on Br^- is used only in the final step, when the bromide ion attacks the carbocation.

QUICK CHECK

Which reagent and observation show that hex-1-ene is unsaturated?

- Bromine water turns from colourless to orange.
- Acidified potassium manganate(VII) turns from colourless to purple.
- Universal indicator turns red.
- Bromine water turns from orange to colourless.

Bromine adds across the C=C, so the orange colour of the bromine disappears. Manganate(VII) also works, but it starts purple and is decolourised, not the other way round.

QUICK CHECK

Which is the repeat unit of the polymer made from chloroethene, CH_2=CHCl?

- $-[CH_2–CHCl–CH_2–CHCl]_n–$
- $-[CH_2–CHCl]_n–$
- $-[CH=CCl]_n–$
- $-[CH_2–CH_2–Cl]_n–$

The repeat unit is just two backbone carbons with the original four atoms (three H, one Cl) still attached, and the double bond has become single. The first option shows two repeat units, the third keeps a double bond, and the last puts the Cl in the chain.

QUICK CHECK

Why can burning waste PVC in an incinerator be a problem, and how is it dealt with?

- It releases chlorine gas, which is removed by passing the gases through water.
- It releases hydrogen chloride, which is removed by reacting it with a base such as calcium oxide.
- It releases sulfur dioxide, which is removed with a catalytic converter.
- It does not burn, so it must go to landfill.

PVC contains chlorine, which ends up as acidic, toxic HCl gas when it burns. Passing the waste gases through a base neutralises it: $CaO + 2HCl \rightarrow CaCl_2 + H_2O$.